SENSI EEG PREPROC

ICA-EKG-TRIGGER

USER MANUAL

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CHAPTER 1

Introduction

Independent Component Analysis (ICA) is widely used in EEG preprocessing to separate neural activity from non-neural contamination, enabling artifacts to be removed while retaining other activity in the data record by omitting identified artifactual components and back-projecting the remaining sources [1, 2]. In practice, however, ICA-based cleaning often remains a manual bottleneck: users must inspect many components and decide which are artifacts, a process that is time-consuming and sometimes subjective.

This Module targets two artifact families that may appear in ICA component space and are well-suited to frequency-guided candidate ranking: cardiac (EKG) contamination and DIN-related periodic interference. EKG contamination can appear as quasi-periodic structure in the time domain and as a harmonic series in the frequency domain; it is a well-documented EEG artifact and has been explicitly studied in ICA-based artifact removal contexts [2].

DINs (Digital Inputs) are widely used in EEG experiments to record event timing and synchronize external devices. In Electrical Geodesics, Inc. (EGI)/Net Station workflows, DIN events are conveyed to the amplifier through a DIN cable/patch cable and are registered as Digital Input events; DINs can be delivered as TTL-style 5V digital signals connected to the amplifier's digital inputs [3]. TTL is a common interfacing standard for EEG amplifiers and is used across multiple acquisition ecosystems (e.g., EGI, Brain Products, and Neuroscan) [4, 5]. Although DIN-related contamination is not a common EEG artifact in the same sense as EKG, we have observed that DIN trigger pulses (or their electrical coupling through cabling/-

grounding) can sometimes manifest as stereotyped transients and/or narrowband comb-like structure in sensor-space EEG spectra, and that this activity tends to be captured in one or a few ICA components. This Module provides a structured, semi-automated workflow to identify such components when present, while preserving user control over final decisions.

1.1 Module Background and Design Motivation

This EKG-DIN Artifact Detection Module was developed as part of the Stanford Educational Neuroscience Initiative (SENSI) EEG preprocessing framework, *SENSI EEG PREPROC*, with the explicit goal of reducing the manual burden of post-ICA artifact cleaning while improving reproducibility. Rather than requiring users to scroll through many components and make ad-hoc decisions, the Module performs automated candidate scoring and presents a small, ranked set of candidates for confirmation in interactive review interfaces. The review step is intentionally retained because artifact morphology can vary across subjects and sessions, and fully automatic rejection rules can over-remove valid components or miss subtle but meaningful contamination.

A second design motivation is documentation. ICA cleaning decisions are often difficult to audit after the fact. The review UIs are designed to (i) keep/remove decisions and (ii) optionally save figures that capture the final decisions, supporting transparent reporting and reanalysis.

This module is part of a larger data-cleaning pipeline (in preparation), which includes additional steps to identify artifactual ICA components, such as those capturing eye movements [6].

1.2 Conceptual Overview of the Module

ICA components are typically evaluated using three complementary views: the scalp projection (topography), the component time course, and the frequency spectrum [7]. This Module leverages that same triad, but uses frequency-domain structure as the primary signal for automated candidate ranking.

For EKG detection, the Module searches for a physiological heart-rate fundamental and evaluates whether the component exhibits energy concentrated at integer harmonics of that fundamental (consistent with cardiac periodicity). For DIN detection, the Module uses the known DIN repetition period to define an expected harmonic spacing and computes a harmonic-energy ratio over a user-set frequency band. In both cases, automated scoring is treated as prioritization: final classification is confirmed in interactive review UIs that display topography, a representative time segment, and the spectrum for each candidate component.

1.3 Scope and Intended Use

This Module is intended as a post-ICA quality-control step in EEG preprocessing. It is designed to be applied after basic filtering and bad-channel handling, once ICA decomposition has been computed, and before downstream analyses.

Default parameter values are chosen to be broadly compatible with typical EEG recordings. In most cases, users should not need to modify these settings. However, some parameters may require adjustment when recording characteristics differ substantially from standard conditions (e.g., unusually low sampling rates, strong downsampling, or nonstandard frequency content). In particular, frequency-domain settings (analysis bands and harmonic window widths) should remain compatible with the sampling rate and the frequency resolution of the spectral estimates.

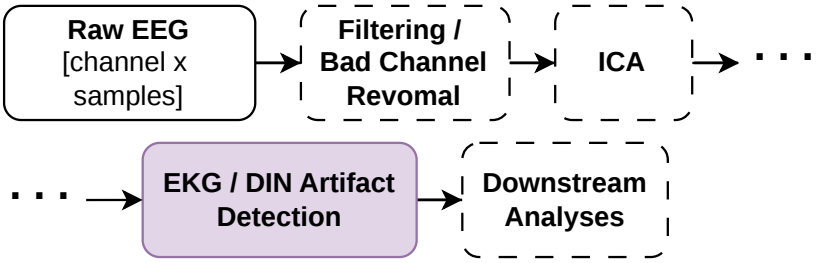


Figure 1: **Intended use.**

Design Philosophy

The central design principle of this module is **guided efficiency**: automated metrics reduce the search space and the interactive interface minimizes code modification while preserving user control. This balance allows the preprocessing workflow to scale to large datasets while reducing manual effort and the likelihood of accidental errors.

1.4 Acknowledgments

The authors thank members of SENSI and collaborating labs at Stanford University for their feedback and early testing of the Module.

This User Manual was prepared using the *Extensive LaTeX Guide* Overleaf template by Armin Dubert.[1]

Author Information and Contributions

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[1] https://www.overleaf.com/articles/extensive-latex-guide/vdgrdqdwjhtk

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Both authors approve the release of the code package and contributed as follows:

- Designed the algorithm and implemented the code: AJM.
- Validated and integrated into larger *SENSI EEG PREPROC* pipelines: AJM, BK.
- Documented the Module: AJM.
- Created illustrative analyses: AJM.
- Wrote the preprint: AJM.
- Provided supervision, feedback, and editing: BK.

Support

For questions, comments, or feature requests, please contact Amilcar Malave <amilcar.malave1997@gmail.com> and Blair Kaneshiro <blairbo@stanford.edu>.

Declaration on the Usage of AI

The authors used OpenAI's GPT-5 and Google's gemini-2.5-flash models for assistance with documentation formatting, function summaries, LaTeX structure, and for brainstorming and exploring conceptual ideas. Additionally, these tools were used to improve flow, word choice, and structure of the User Manual and preprint. No AI tools were used for data analysis, algorithm development, evaluation, or initial drafts of the User Manual or preprint.

License

SENSI EEG PREPROC ICA-EKG-Trigger Detection Module is released under the MIT License,[2] as follows:

```
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```

[2] https://choosealicense.com/licenses/mit/

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If a code file within the toolbox originated from outside of "*SENSI EEG PREPROC* ICA-EKG-Trigger Module" and already contained a license, that original license is retained.

Citing the Module

The ICA-EKG-Trigger Module release comprises three deliverables: the codebase, example data [8], and a preprint summarizing the release [9].

If using the ICA-EKG-Trigger Artifact Detection Module or the associated figures or methods, please cite the following:

- **GitHub repository**
  Amilcar J. Malave & Blair Kaneshiro (2026). SENSI-EEG-Preproc-ICA-EKG-Trigger (v1.0): A MATLAB framework for semi-automated identification of EKG and trigger artifacts in EEG using ICA and spectral characteristics. Stanford University.
  https://github.com/edneuro/SENSI-EEG-Preproc-ICA-EKG-Trigger.
  https://doi.org/10.25740/pm515wq0875

- **Release Preprint**
  Amilcar J. Malave & Blair Kaneshiro (2026). Semi-Automated Identification of EKG and Trigger Artifacts in EEG Using ICA and Spectral Characteristics. bioRxiv.
  https://doi.org/10.64898/2026.04.08.717297

If using the example data accompanying this software release in other projects, please cite the following:

- **Dataset**
  Amilcar J. Malave & Blair Kaneshiro (2026). Example Data for SENSI-EEG-Preproc-ICA-EKG-Trigger [Data set]. Stanford Digital Repository.
  https://doi.org/10.25740/xd818jt7842

CHAPTER 2

Getting Started

2.1 Toolbox and Resources

2.1.1 GitHub Repository

The *SENSI EEG PREPROC* EKG-DIN Artifact Detection Module is available as a publicly accessible GitHub repository.[3] The citation for the repository is associated with the linked instance on the Stanford Digital Repository. As noted in Chapter 1.4, this Module is published under an MIT license.

2.1.2 User Manual and Preprint

The *SENSI EEG PREPROC* EKG-DIN Artifact Detection Module release includes two forms of written documentation.

First, the User Manual (this document) is available in the GitHub repository. It offers comprehensive documentation of the Module, including a brief background on EKG and DIN artifacts, guidance on getting started, a walkthrough of the code, illustrative analyses, and detailed documentation of the main and helper functions.

The preprint of the release is available on bioRxiv [9]. It provides a more narrative overview of the Module's structure, functions, and illustrative analyses.

[3] https://github.com/edneuro/SENSI-EEG-Preproc-ICA-EKG-Trigger

2.1.3 Example Data

Due to file size, example data files used in the illustrative analyses are not included on this Module's repository on GitHub and hence this folder contains only a `readme.txt` file. They are instead provided as a separate dataset through the Stanford Digital Repository (SDR) [8].[4]

The example script in the main folder of the repository **`(example.m)`** downloads the data files from their SDR URLs into the **ExampleData** folder in the user's local instance of "*SENSI EEG PREPROC* EKG-DIN Artifact Detection Module". More information on the example data files is provided in the README of the online dataset.

For more information, the downloading procedure is provided in Section 5.2.

2.2 Setup and Installation

2.2.1 Operating Systems and MATLAB versions

This Module was validated on both Windows and Linux operating systems, and we anticipate that the current release will work similarly across operating systems, but this has not been comprehensively validated.

This Module was developed and tested on MATLAB R2024b. The Module may work with earlier versions of MATLAB, but it has not been fully tested on previous versions.

2.2.2 Dependencies

The software requires two MATLAB toolboxes:

- Signal Processing Toolbox[5].
- Statistics and Machine Learning Toolbox[6].

> Run the following command in MATLAB to determine whether the Toolbox is installed:
>
> ```
> >> ver
> ```
>
> This lists all installed toolboxes.

2.2.3 Installation

Clone or download the latest version of the SENSI EEG PREPROC ICA-EKG-Trigger Detection Module at: `https://github.com/edneuro/SENSI-EEG-Preproc-ICA-EKG-Trigger`

[4] `https://purl.stanford.edu/xd818jt7842`
[5] `https://www.mathworks.com/products/signal.html`
[6] `https://www.mathworks.com/products/statistics.html`

After downloading the codebase, the user can run the following in the MATLAB command window to add the entire Module to their **path**:

```
> CODEBASE_PATH = 'path/to/your/directory/in/char/format';
> addpath(genpath(CODEBASE_PATH));
```

Some of the path specifications in **example.m** are relative to the current working directory. If absolute folder paths are not specified (described later), the user must ensure that MATLAB is operating from the Module's main folder.

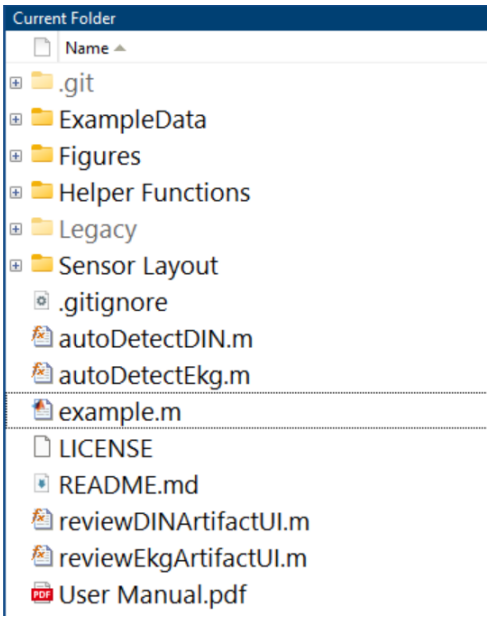


Figure 2: **Current folder.** Ensure that the MATLAB current folder is set to the module directory when running `example.m`.

2.3 Using the Module

This Module provides a post-ICA workflow for identifying ICA components contaminated by (i) cardiac (EKG) artifacts and (ii) DIN-related periodic interference. The intended usage pattern is:

1. run automated candidate detection on ICA components,
2. confirm decisions in the interactive review interfaces,
3. remove confirmed artifact components and reconstruct cleaned sensor-space data.

This Module is function-based and does not include a full graphical interface. All operations

are carried out through MATLAB functions, with interactive review user interfaces (UI) used only for the final decisions. Users should be comfortable calling functions from the MATLAB command window or from their own preprocessing scripts.

2.3.1 Folder Overview

The top-level folder of the Module contains the following directories and files:

- ***ExampleData/*** — Contains only a `readme.txt` file by default. Example datasets used by `example.m` are downloaded into this folder automatically.
- ***Figures/*** — Default folder used by `example.m` to save generated figures.
- ***HelperFunctions/*** — Contains all helper functions used by `autoDetectEKG() & autoDetectDIN()`. These functions are not typically called directly by the user and are documented separately in Section 7.
- ***Sensor Layouts/*** — Contains sensor-layout–specific `chanlocs` and related layout files used for topographic plotting. Details on file formats are provided in Section 2.3.4.
- ***example.m*** — A complete example script showing how to configure the Module, run `autoDetectEKG() & autoDetectDIN()`, and inspect the outputs. This is the recommended starting point for new users.
- ***autoDetectEKG.m*** — Main function for detecting EKG components in ICA space.
- ***autoDetectDIN.m*** — Main function for detecting DIN components in ICA space.
- ***reviewEKGArtifactUI.m*** — EKG interactive review and final source selection.
- ***reviewDINArtifactUI.m*** — DIN interactive review and final source selection.
- ***README.md*** — A brief overview of the Module.
- ***LICENSE*** — License for the repository (MIT license).
- **User Manual.pdf** (this file) — User Manual for the current release.
- ***.gitignore*** — Git configuration (no user action required).

2.3.2 Main Functions

This Module is designed to be used through four main functions: two automated detectors and two interactive review interfaces.

Automated detectors.

`autoDetectDIN()` — Computes a DIN harmonic-ratio score for each ICA component and returns a small set of candidate DIN components for review.

```
[dinSrc, dinInfo] = autoDetectDIN(xICA, fs, nSources, dinOpts)
```

`autoDetectEkg()` — Computes an EKG harmonic local-SNR score for each ICA component and returns (i) strong EKG candidates and (ii) additional suspects for review.

```
[ekgSrc, ~, ~] = autoDetectEkg(xICA, fs, nSources, ekgOpts)
```

Interactive review interfaces.

`reviewDINArtifactUI()` — UI for confirming DIN candidates. Returns the final DIN component list.

```
dinSrc = reviewDINArtifactUI(W, xICA, fs, chanlocs, dinSrc, ...
    dinInfo.candidates, StartSec, NSec, figName, saveFigs, saveFolder);
```

`reviewEkgArtifactUI()` — UI for confirming EKG candidates. Returns the final EKG component list.

```
ekgSrc = reviewEkgArtifactUI(W, xICA, fs, chanlocs, ekgSrc, ...
    ekgSus, StartSec, NSec, figName, saveFigs, saveFolder)
```

2.3.3 Inputs and Outputs

All functions operate on variables already loaded into the MATLAB workspace.

Shared Inputs

- `xICA` — ICA component time series of shape [components × samples], computed as `xICA` = $W \cdot$ `xRaw`.
- `fs` — Sampling rate in Hz (e.g., 250).
- `W` — ICA unmixing matrix used to compute ICA sources and to render topographies.
- `chanlocs` = chanlocsFromFile('EGI-9_18-124.sfp') — Electrode location structure.
- `nSources` = 1:size(xICA,1) — Vector of component indices to scan.
- `StartSec` = 300 — Where to start review time plots.
- `NSec` = 10 — How many seconds to show in review time plots.
- `saveFigs` = 1 — Controlling whether review figures are saved.
- `saveFolder = 'Figs'` — Output folder path for saved figures.
- `figName` = 'data1_DIN'— Base filename to save figures.

`autoDetectEkg()`

Inputs.

- `ekgOpts` — Configuration struct controlling harmonic local-SNR scoring.
  - `ekgOpts.fmin = 0.8;` (minimum heart-rate frequency in Hz)
  - `ekgOpts.fmax = 2.0;` (maximum heart-rate frequency in Hz)

  Other fields may be left at defaults (harmonics, windows, detrending, Welch length, ...)

Outputs.

- `ekgSrc` — Strong EKG candidates returned for interactive confirmation.
- `ekgSus` — Additional suspects returned for review.

`reviewEkgArtifactUI()`

- `ekgSrc` — Final EKG component list confirmed by the user.

`autoDetectDIN()`

Inputs.

- `dinOpts` — Configuration struct controlling DIN harmonic-ratio computation.
  - `dinOpts.T = 1;` (DIN repetition period in seconds)
  - `dinOpts.fmin = 10;` (minimum frequency for analysis in Hz)
  - `dinOpts.fmax = 50;` (maximum frequency for analysis in Hz)

  Other fields may be left at defaults (bg, guard, bwFrac, method, ...)

Outputs.

- `dinSrc` — Strong DIN candidates returned for interactive review
- `dinInfo.candidates` — Weak DIN candidates.

`reviewDINArtifactUI()`

Outputs.

- `dinSrc` — Final DIN component list confirmed by the user.

2.3.4 Sensor Layouts

This project uses sensor layout files to define electrode coordinates for topography plotting (e.g., ICA scalp maps shown in `reviewDINArtifactUI()` and `reviewEkgArtifactUI()`).

Location of sensor layout files. Sensor layout files are stored in:

```
Sensor Layouts/
```

How sensor layouts are used. Sensor layouts are loaded using `chanlocsFromFile()`, which parses an electrode-location file and returns a `chanlocs` struct compatible with topoplot functions:

```
chanlocs = chanlocsFromFile('EGI-9_18-124.sfp');
```

The `chanlocs` output is then passed to the review UIs for rendering ICA component scalp maps.

Supported file formats. `chanlocsFromFile()` supports multiple common electrode-location formats, including: `polhemus`, `besa`, `xyz`, `sfp`, `loc`, `sph`, `asc`, and others. Additional custom formats may be added directly within `chanlocsFromFile()`.

File requirements. Sensor layout files must contain **only electrode entries** (no header text, metadata blocks, or reference channel rows). Each row must correspond to exactly one electrode present in the input data matrix `xRaw`. If the dataset has removed channels (e.g., `badCh`), the corresponding entries must also be removed from the `chanlocs` structure to preserve index alignment.

If bad channels were removed from `xRaw` (e.g., via `badCh`), the corresponding rows **must also be removed** from `chanlocs` before passing it to the review UIs — mismatched sizes will cause incorrect topography rendering. This step is automatically done in the `example.m`

Example: EGI `.sfp` file. For example, the file `EGI-9_18-124.sfp` contains 124 electrode rows (one per channel in `xRaw`), with the heading and Cz removed:

```
E1      6.156820   4.490185  -3.420040
E2      5.871912   5.885035  -0.413980
E3      4.780578   7.072385   1.994388
...
```

```
E123    6.282423    4.671705    1.732265
E124    5.138550    5.716433    3.445657
```

Available sensor layouts. Currently included layouts:

EGI 9–18 years old.

- `EGI-9_18-128.sfp` [10]
- `EGI-9_18-124.sfp` (derived from `EGI-9_18-128.sfp`)

Using a different sensor layout. If a user employs a montage not included in `Sensor Layouts/`, they must provide a valid location file in a supported format, ensure it contains only electrode rows (no headers or reference channel rows), and confirm that the number and ordering of electrodes match the channel ordering in `xRaw`. If needed, new formats may be added to `chanlocsFromFile()`.

CHAPTER 3

autoDetectEkg.m reviewEkgArtifactUI.m

Structure of this chapter.

This chapter is organized as follows:

1. Overview (Section 3.1).
2. Definition of the function and its inputs (Section 3.2).
3. Mathematical formulation of the harmonic local-SNR metric (Section 3.3).
4. Interactive review and final confirmation (Section 3.4).
5. Discussion of alternative strategies and design decisions (Section 3.5).

3.1 Overview

This chapter describes the harmonic signal-to-noise ratio (SNR)–based procedure used to detect ICA components contaminated by cardiac (EKG) artifacts. The method operates in the frequency domain and evaluates whether a component exhibits spectral energy concentrated at integer multiples of a physiological heart-rate frequency.

Detection is fully automated at the scoring stage, but final rejection decisions are made through an interactive review interface (`reviewEkgArtifactUI()`).

Data assumptions. This detector is designed for ICA component time series computed from EEG data that have already undergone bandpass filtered (high-pass and low-pass) and removal of bad channels.

Figure 3 illustrates a representative EKG component, including its topography, 10 seconds of time-domain waveform, and magnitude spectrum.

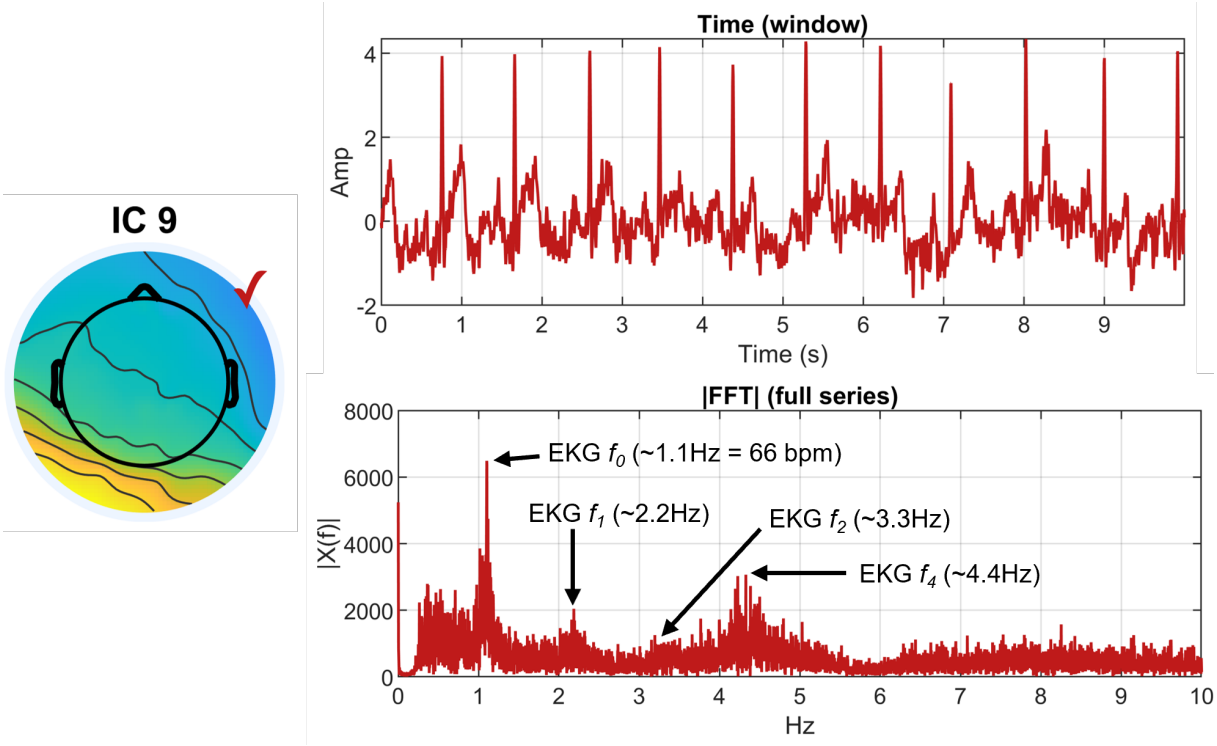


Figure 3: **Representative EKG ICA component.** *Left:* Spatial topography. *Top-right:* Time-domain waveform. *Bottom-right:* Frequency domain.

Explored detection strategies. We evaluated multiple approaches for automated EKG detection:

1. **Time-domain periodicity detection:** identifying quasi-periodic peaks directly in the component waveform.
2. **Topographic heuristics:** identifying components whose scalp projections are broadly distributed
3. **Frequency-domain harmonic structure detection:** quantifying the presence of heart-rate frequency harmonics.

The current release of the Module ultimately uses only option 3 from the above list. Discussion of options 1 and 2 are described in Section 3.5.

Cardiac contamination consistently produces a fundamental frequency within the physiological heart-rate range (approximately 0.8–2.0 Hz), accompanied by harmonics extending across the spectrum. While waveform morphology and spatial projection can vary, this harmonic structure remains stable across subjects and sessions.

3.2 autoDetectEkg()

Syntax

```
[ekgSrc, ekgSus, metrics] = autoDetectEkg(xICA, fs, nSources, opts)
```

Description. This is the main entry point of the EKG artifact detection portion of the Module. It scores ICA components using harmonic local-SNR: it searches for a candidate fundamental frequency in a physiological band, evaluates integer harmonics of that frequency, and quantifies how strongly each harmonic stands out relative to its local spectral neighborhood. The resulting harmonic scores are aggregated into a composite score that can be used to prioritize components for visual inspection.

Inputs.

- `xICA`: ICA component time series, shape $[n_{\text{Comp}} \times n_{\text{Samp}}]$. Each row corresponds to one ICA component.
- `fs`: Sampling rate (Hz).
- `nSources`: Vector of component indices to scan. If empty, all components are scanned.
- `opts`: Options structure controlling spectral estimation and harmonic scoring. Key fields (defaults shown) include:
  - `window` (Default = 14) | Welch window length (sec).
  - `fmin` (Default = 0.8), `fmax` (Default = 2.0) | fundamental search band (Hz).
  - `harmonics` (Default = 4) | maximum number of harmonics to evaluate.
  - `peakHalfHz` (Default = 0.2) | half-width of harmonic peak window (Hz).
  - `nbHalfHz` (Default = 0.6) | half-width of neighbor band around each harmonic (Hz).
  - `do_detrend` (Default = 1): enable log–log spectral detrending (1/f flattening).
  - `detrend_fmin` (Default = 0.5), `detrend_fmax` (Default = 30) | fitting band for spectral detrending (Hz).

3.3 autoDetectEkg() Processing Flow

The EKG detection procedure consists of five stages:

1. Power spectral density estimation via Welch's method.
2. Identification of a candidate fundamental frequency within a physiological band.
3. Harmonic SNR computation at integer multiples of the fundamental.
4. Aggregation of harmonic scores into a composite metric.
5. Robust standardization across components and candidate selection.

3.3.1 Step 1: Spectral Estimation

For each ICA component $c \in \{1, \ldots, n_{\text{Comp}}\}$, the time series $x_c(t)$ is converted to the frequency domain using Welch's method:

$$P_c(f) = \texttt{pwelch}\big(x_c(t)\big),$$

where the window length and overlap are defined by `opts.window` and `opts.overlap`, respectively.

If enabled (`opts.do_detrend` = 1), the log–log spectrum is linearly detrended over the range [`opts.detrend_fmin`, `opts.detrend_fmax`] to reduce broadband $1/f$-like structure prior to harmonic evaluation. This step flattens the spectrum to emphasize narrowband periodic structure.

The resulting power spectrum $P_c(f)$ and frequency axis are stored in `metrics.psdPow` and `metrics.psdFreq`.

3.3.2 Step 2: Fundamental Frequency Identification

Cardiac contamination is expected to produce a fundamental frequency within a physiological band:

$$f_1 \in [f_{\min}, f_{\max}],$$

where defaults are

$$f_{\min} = 0.8 \text{ Hz}, \quad f_{\max} = 2.0 \text{ Hz}.$$

For each component, the frequency within this band with maximal spectral power is selected:

$$f_{1,c} = \arg \max_{f \in [f_{\min}, f_{\max}]} P_c(f).$$

This value is stored as `metrics.fPeak`.

3.3.3 Step 3: Harmonic Local-SNR Computation

Given the detected fundamental frequency $f_{1,c}$, expected harmonic frequencies are defined as:

$$f_{h,c} = h \cdot f_{1,c}, \qquad h = 1, 2, \ldots, H,$$

where $H =$ `opts.harmonics` (default $H = 4$).

Peak and neighbor frequency regions. For each harmonic $f_{h,c}$, two frequency regions are defined:

1. A narrow *peak region* centered at the harmonic frequency:

   $$\mathscr{P}_{h,c} = \left[f_{h,c} - \Delta_p,\ f_{h,c} + \Delta_p \right],$$

   where $\Delta_p =$ `opts.peakHalfHz` defines the half-width (in Hz) of the harmonic peak window.

2. A surrounding *neighbor region* used to estimate the local spectral baseline:

   $$\mathscr{N}_{h,c} = \left(f_{h,c} - \Delta_n,\ f_{h,c} - \Delta_p \right) \cup \left(f_{h,c} + \Delta_p,\ f_{h,c} + \Delta_n \right),$$

   where $\Delta_n =$ `opts.nbHalfHz` defines the outer half-width of the surrounding baseline region.

The peak region captures harmonic energy, while the neighbor region captures local background power immediately adjacent to the harmonic.

Local harmonic SNR. Let $P_c(f)$ denote the power spectral density (PSD) of component c. Peak power is computed as:

$$S_{h,c}^{(\text{peak})} = \sum_{f \in \mathscr{P}_{h,c}} P_c(f).$$

Neighbor power is computed analogously:

$$S_{h,c}^{(\text{neighbor})} = \text{median}_{f \in \mathscr{N}_{h,c}} P_c(f).$$

The harmonic local SNR is then defined as:

$$\mathrm{SNR}_{h,c} = \frac{S_{h,c}^{(\mathrm{peak})}}{S_{h,c}^{(\mathrm{neighbor})}}.$$

All harmonic SNR values are stored in `metrics.harmSNR`.

3.3.4 Step 4: Composite Harmonic Score

For each component, harmonic SNR values are aggregated:

$$\mathrm{HarmonicSNRsum}_c = \sum_{h=1}^{H} \mathrm{SNR}_{h,c}.$$

The intuition is that true cardiac contamination produces not only a strong fundamental, but a structured harmonic series. Summing across harmonics increases robustness to variability in any single harmonic peak.

3.3.5 Step 5: Robust Scoring and Candidate Selection

This stage converts `harmSNRsum` into a robust, cross-component outlier score and uses it to generate two ranked output lists: `ekgSrc` and `ekgSus`. Importantly, the function returns at most three total components across both lists.

Robust composite score (median/MAD z-score)

Let v_c denote the harmonic composite score for component c,

$$v_c = \mathrm{harmSNRsum}_c.$$

A robust location and scale are computed across the scanned components in `nSources`:

$$m = \mathrm{median}(v_c), \qquad \mathrm{MAD} = 1.4826 \cdot \mathrm{median}\,|v_c - m|,$$

The robust composite score is then:

$$z_c = \frac{v_c - m}{\mathrm{MAD}}.$$

These values are stored in `metrics.compositeScore`.

Candidate ranking and capped selection

Components with $z_c > 5$ (high-value outliers) are ranked in descending order. These components are assigned to `ekgSrc`, capped at `maxTot` = 3.

If `ekgSrc` contains fewer than three components, remaining slots are filled with the next-highest ranked components ($z_c > 3$), which are returned as `ekgSus`.

At most three total components (`ekgSrc` + `ekgSus`) are returned.

3.4 Interactive Review and Final Source Selection

The final stage is performed by `reviewEkgArtifactUI()`. The interface presents:

- Component topography.
- A representative time-domain segment.
- The full-series magnitude spectrum.

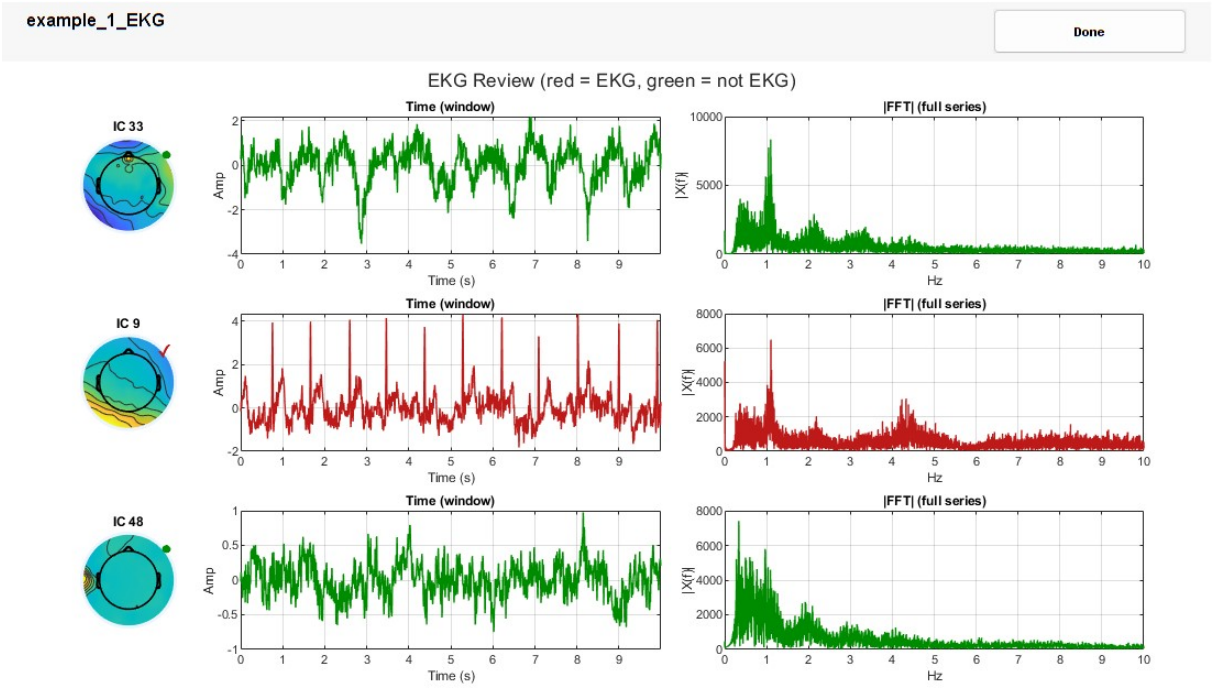


Figure 4: **Interactive EKG review interface.** Each row corresponds to a candidate ICA component. *Left:* Scalp topography. *Middle:* Time-domain waveform. *Right:* Magnitude spectrum highlighting harmonic structure.

Components initially flagged as EKG candidates (`ekgSrc`) are displayed in red (Figure 4). Clicking on a row toggles the classification between:

$$\text{EKG}\,(red = remove) \leftrightarrow \text{Keep}\,(green).$$

The review concludes when the user selects *Done*.



Only components confirmed as EKG at this stage are returned in `ekgSrc`.
Output Rule. `ekgSrc` = {c | user confirms EKG}.
Suspected but unconfirmed components are retained in the data.

3.5 Alternative Strategies and Design Decisions

The final implementation of `autoDetectEkg()` relies on frequency-domain harmonic structure as the primary discriminative feature. During development, two additional strategies were evaluated: time-domain periodicity analysis and topography-based heuristics. Neither proved sufficiently reliable for automated classification.

Time-domain periodicity analysis

One approach considered was identification of repeating cardiac peaks in the ICA time series. Although EKG activity produces periodic beats, the inter-beat interval varies over time due to physiological heart-rate variability. This complicates automated peak-detection rules and reduces the robustness of template-based or interval-consistency metrics.

In contrast, the frequency-domain representation preserves energy concentration near the fundamental heart-rate frequency and its harmonics, even under moderate temporal variability, providing a more stable basis for automated scoring.

Topography-based heuristics

A second strategy considered was classification based on ICA spatial projections (scalp maps). The motivation was that cardiac field activity recorded at the scalp often appears as a broadly distributed pattern with a smooth polarity gradient (i.e., a dipole-like map), consistent with a far-field generator whose electric field propagates through a large volume conductor (thorax/neck/head) before being measured at EEG electrodes [11]. In many recordings, this yields a gradual anterior–posterior or left–right voltage gradient rather than a highly focal pattern.

To test this, we implemented a prototype detector that explicitly scored the presence of a long, smooth dipole-like gradient.

Function Location: `Helper Functions\postICAartifacts\old\autoDetectEkgTopo.m`

For each candidate ICA component, the function: (i) interpolates the component scalp map on a dense grid, (ii) applies a light spatial smoothing and locates the dominant positive and negative poles, (iii) computes the pole-to-pole distance as a “dipole length” measure, and (iv) sampled voltages along the pole axis to quantify whether the map changed smoothly

and monotonically (penalizing abrupt spatial fluctuations). The final score was the product of dipole length and smoothness.

Although this heuristic correctly highlighted many cardiac-related components, it produced systematic false negatives. In particular, some confirmed EKG components did not exhibit a single dominant directional gradient (Figure 5), causing the pole-axis smoothness criterion to fail.

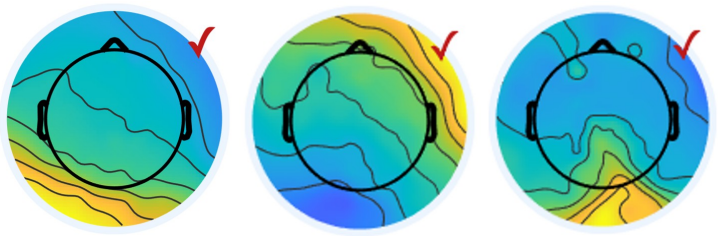


Figure 5: **EKG ICA topographies across recordings.** Each scalp map shows an ICA component identified as cardiac EKG contamination in different recordings. *Left and center*: Components exhibit the expected dipole-like scalp distribution with a smooth polarity gradient. *Right*: A confirmed cardiac component with elevated occipital amplitude, but without a single dominant dipole axis or clear directional gradient, illustrating a failure mode for dipole-gradient heuristics.

For these reasons, topography was retained as a confirmation feature within the manual review interface, but not incorporated into the automated decision rule.

4

CHAPTER

autoDetectDIN.m reviewDINArtifactUI.m

Structure of this chapter.

This chapter is organized as follows:

1. Overview (Section 4.1).
2. Definition of the function and its inputs/outputs (Section 4.2).
3. Mathematical formulation of the DIN harmonic energy ratio metric (Section 4.3).
4. Interactive review and final confirmation (Section 4.4).
5. Discussion of design decisions (Section 4.5).

4.1 Overview

This chapter describes the stimulus-locked harmonic detection procedure used to identify ICA components contaminated by DIN-trigger artifacts. DIN-related contamination is not expected in all EEG recordings; it is system- and setup-dependent and is most relevant in acquisitions where digital input (DIN/trigger) lines are used for synchronization. In our recordings, this artifact has been occasionally observed in EGI/Net Station workflows when DIN triggers were delivered at a periodic rate of 1 Hz.

Unlike physiological artifacts such as EKG, DIN contamination is externally generated and

timed to DIN events. When DIN triggers are sent at a precisely controlled repetition period, affected components exhibit spectral energy concentrated at integer multiples of the known stimulus repetition frequency.

The detection procedure leverages this property by computing a harmonic energy ratio: the proportion of spectral energy located at stimulus-locked harmonics relative to the surrounding baseline spectrum.

As with the EKG detector, scoring is fully automated, but final classification is confirmed through an interactive review interface (`reviewDINArtifactUI()`).

Data assumptions. This detector is designed for ICA component time series computed from EEG data that have already undergone bandpass filtered (high-pass and low-pass) and removal of bad channels.

Figure 6 illustrates a representative DIN component, including its topography, time-domain waveform, and magnitude spectrum.

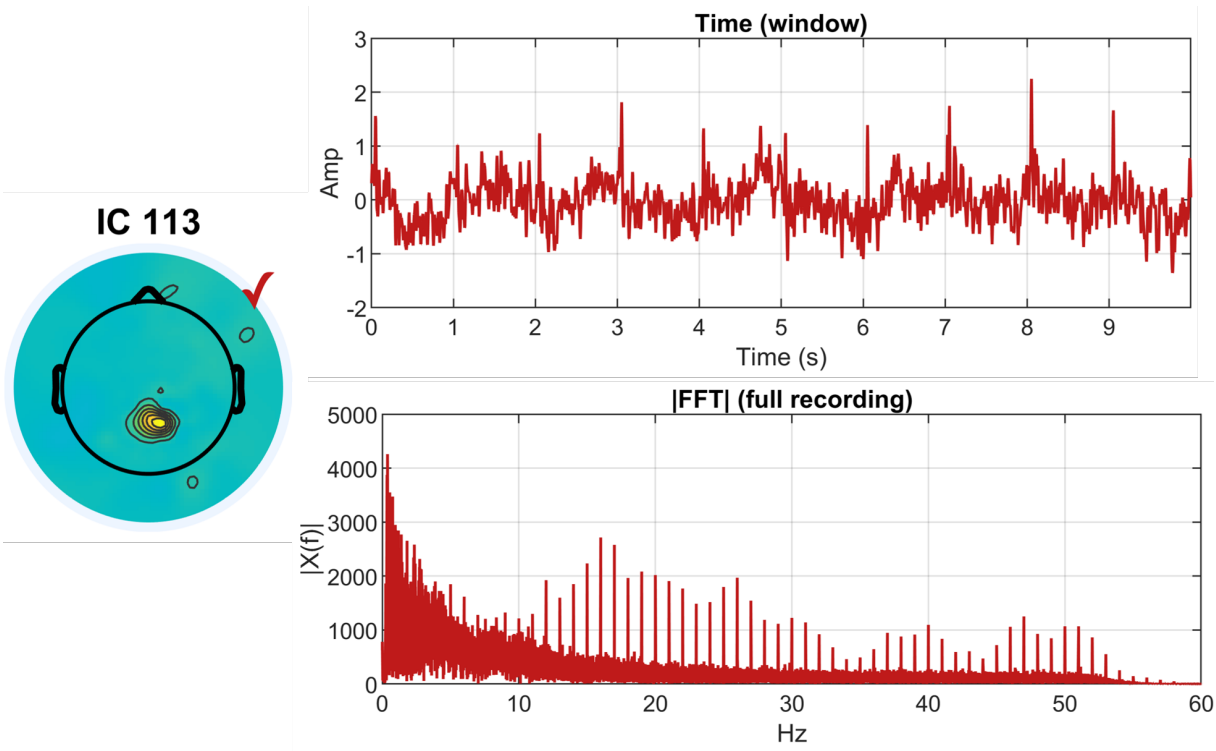


Figure 6: **Representative DIN ICA component.** *Left:* Spatial topography. *Top-right:* Time-domain waveform. *Bottom-right:* Frequency domain.

Explored detection strategies. We evaluated multiple approaches for automated DIN detection:

1. **Time-domain periodicity detection:** identifying periodic peaks directly in the component waveform.
2. **Topographic heuristics:** identifying components whose scalp projections implicate the COM electrode region of the EGI system
3. **Frequency-domain harmonic structure detection:** quantifying the presence of DIN comb frequency harmonics.



The current release of the Module ultimately uses only option 3 from the above list. Discussion of options 1 and 2 are described in Section 4.5.

4.2 autoDetectDIN()

Syntax

```
[dinSrc, info] = autoDetectDIN(xICA, fs, nSources, Opts)
```

Description. `autoDetectDIN()` scans the ICA time series and computes a DIN metric for each component (using `harmRatioFromSignal()`). The resulting scores are standardized across the selected components and thresholded to identify likely DIN sources. The function returns up to three total components as `dinSrc`, and also returns an `info` struct containing all intermediate metrics used for ranking and review.

Inputs.

- `xICA`: ICA component time series, shape $[n_{\text{Comp}} \times n_{\text{Samp}}]$. Each row corresponds to one ICA component.
- `fs`: Sampling rate (Hz).
- `nSources`: Vector of component indices to scan. If empty, all components are scanned.
- `opts`: Options structure controlling spectral estimation and harmonic scoring. Key fields (defaults shown) include:
  - `T` (Default = 1) | DIN repetition period (sec).
  - `fmin` (Default = 10), `fmax` (Default = 50) | analysis band (Hz).
  - `bwFrac` (Default = 0.15) | bandwidth fraction of f_0 (used to compute harmonic peak window).
  - `guardFrac` (Default = 0.08) | guard band fraction of f_0 (used to compute neighbor baseline boundaries).
  - `method` (Default = 'welch') | method for spectrum estimation ('welch' | 'fft' | 'pmtm').

- segLen | segment length for Welch (samples); defaults to 14–30 seconds of data.
- overlapFrac (Default = 0.5) | Welch overlap fraction.

4.3 autoDetectDIN() Processing Flow

1. Compute DIN metric per component using harmRatioFromSignal().
2. z-score DIN metrics across nSources and select candidates.
3. Interactive review and final selection via reviewDINArtifactUI().

4.3.1 Step 1: Computation of harmRatioFromSignal()

DIN contamination produces a narrowband periodic spectral structure. Rather than searching for physiological fundamentals (as in EKG detection), the DIN detector evaluates whether a component exhibits a harmonic energy pattern consistent with the digital interference signal.

Let $x_c(t)$ denote the ICA time series for component c. The first step is spectral estimation using Welch's method:

$$P_c(f) = \texttt{pwelch}(x_c(t)),$$

where $P_c(f)$ is the one-sided power spectral density (PSD).

Harmonic frequency set. The DIN metric is parameterized by the DIN repetition period T_{DIN} (seconds), provided via the options structure. The corresponding base frequency is:

$$f_0 = \frac{1}{T_{\text{DIN}}}.$$

Harmonic frequencies are defined as integer multiples of f_0:

$$f_h = h \cdot f_0, \qquad h \in \mathbb{N}.$$

However, harmonics are included only if they fall within the valid spectral range:

$$f_{\min} \le f_h \le f_{\max},$$

where $f_{\min}$ and $f_{\max}$ define the frequency interval over which the harmonic ratio is evaluated.

Thus, the harmonic index set is:

$$\mathcal{H} = \{h \in \mathbb{N} \mid f_{\min} \le h f_0 \le f_{\max}\}.$$

The number of harmonics is therefore determined automatically by the spectrum and the DIN repetition period.

Peak regions (harmonic energy). For each harmonic f_h, a narrow peak window is defined:

$$\mathcal{P}_h = [f_h - \Delta_p,\ f_h + \Delta_p],$$

where Δ_p is a small half-width (Hz).

Harmonic energy is computed by summing spectral power within these windows:

$$E_{\text{harm}} = \sum_{h \in \mathcal{H}} \sum_{f \in \mathcal{P}_h} P_c(f).$$

This measures the total spectral energy concentrated at the expected harmonic frequencies.

Baseline (local spectral neighborhood). To estimate background spectral energy, a local spectral neighborhood is defined for each harmonic:

$$\mathcal{N}_h = (f_h - \Delta_n,\ f_h - \Delta_p) \cup (f_h + \Delta_p,\ f_h + \Delta_n),$$

where $\Delta_n > \Delta_p$.

The baseline energy is computed as:

$$E_{\text{base}} = \sum_{h \in \mathcal{H}} \text{median}_{f \in \mathcal{N}_h} P_c(f).$$

The median is used to obtain a robust estimate of local background power, preventing isolated peaks from inflating the baseline estimate.

Harmonic ratio. The harmonic ratio returned by the function is:

$$R_c = \frac{E_{\text{harm}}}{E_{\text{base}}},$$

Large values of R_c indicate that spectral energy is strongly concentrated at integer multiples of f_0, consistent with structured digital interference.

4.3.2 Step 2: Cross-Component Standardization

The harmonic-ratio metric R_c is interpreted relatively across ICA components. DIN components are expected to appear as *positive outliers* within the population of scanned sources.

Let $\mathcal{S}$ denote the set of scanned components (indices in `nSources`):

$$z_c = zscore(R_c), \qquad c \in \mathcal{S}.$$

Large positive z_c indicates that harmonic energy is substantially stronger than the component population baseline.

4.3.3 Step 3: Two-Stage Candidate Selection (Maximum 3 Components)

Candidate selection is formulated as an outlier-ranking problem with a hard cap of three total components.

Stage 1 (primary outliers). A fixed threshold $\tau = 2.5$ is applied:

$$\mathcal{C}_1 = \{c \in \mathcal{S} \mid z_c > \tau\}.$$

Components in $\mathcal{C}_1$ are sorted in descending order of z_c. The strongest components (up to three) are assigned to **`dinSrc`**.

Stage 2 (secondary pass). If fewer than three components exceed the primary threshold, additional high-ranking components (after re-standardization) are added to `info.reviewCandidates` for interactive review.

Interpretation. DIN detection is a cross-component outlier detection problem in harmonic-ratio space. The two-stage procedure ensures that a dominant outlier does not mask additional DIN components, while maintaining a strict maximum of three candidates.

4.4 Interactive Review and Final Source Selection

The final stage is performed by `reviewDINArtifactUI()`. The interface presents:

- Component topography.
- A representative time-domain segment.
- The full-series magnitude spectrum.

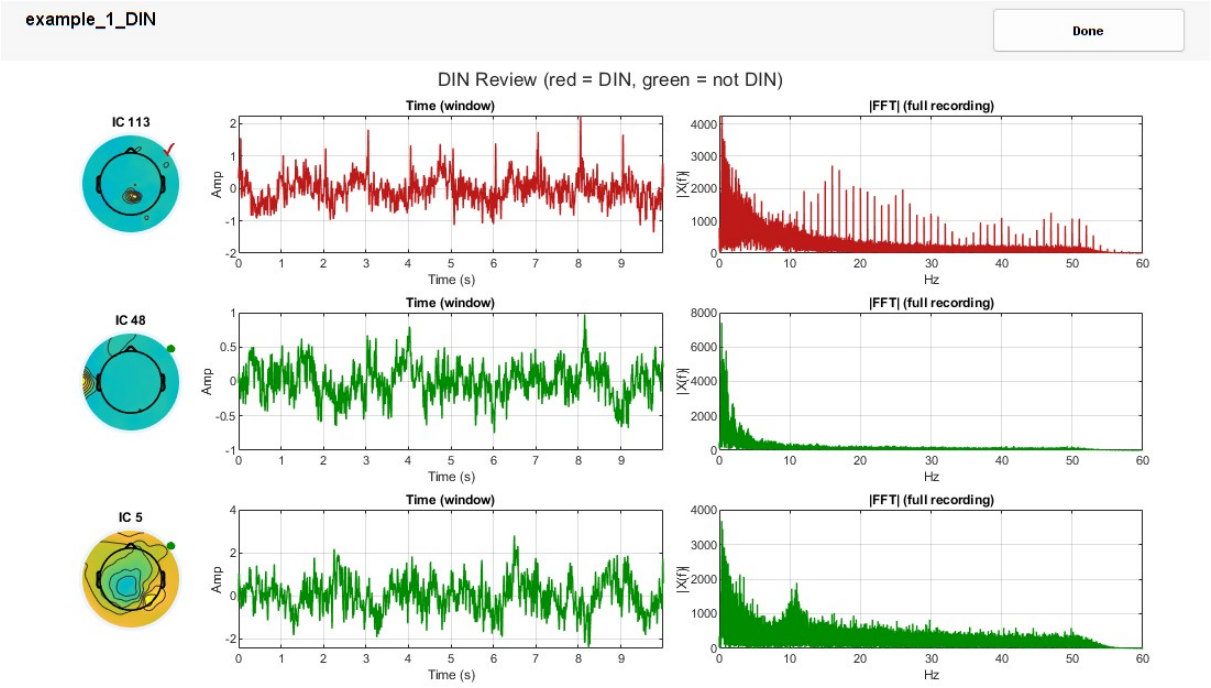


Figure 7: **Interactive DIN review interface.** Each row corresponds to a candidate ICA component. *Left:* Scalp topography. *Middle:* Time-domain waveform. *Right:* Magnitude spectrum highlighting harmonic structure.

Components initially flagged as DIN candidates (`dinSrc`) are displayed in red (Figure 7). Clicking on a row toggles the classification between:

$$\text{DIN } (red = remove) \leftrightarrow \text{Keep } (green).$$

The review concludes when the user selects *Done*.

Only components confirmed as DIN at this stage are returned in `dinSrc`.
Output Rule. `dinSrc` = $\{c \mid \text{user confirms DIN}\}$.
Suspected but unconfirmed components are retained in the data.

4.5 Alternative Strategies and Design Decisions

The final implementation of `autoDetectDIN()` relies on the frequency-domain signature of the artifact as the primary discriminative feature. During development, two additional strategies were evaluated: time-domain periodicity analysis and topography-based heuristics.

Time-domain periodicity analysis

A natural first approach was to exploit the distinctive time-domain morphology of DIN contamination in the ICA activations. In the time series, DIN components often contain brief pulses that recur at fixed intervals. In principle, this structure enables a matched-filter (template) detector, where a representative transient waveform could be used identify ICA component with high template correlation.

In contrast, the frequency-domain representation provided an immediately distinctive and consistent signature. DIN components exhibited a pronounced comb-like spectrum with strong energy at a fundamental frequency and its harmonics. This harmonic "line" structure made DIN identification substantially easier to automate than time-domain template matching.

Topography-based heuristics

A second strategy considered was classification based on ICA scalp maps. In many recordings, DIN-related components projected as a highly focal pattern, often recurring in a similar region across sessions, suggesting a spatially localized source. This motivated an attempted heuristic that would flag components based on the presence of a strong local maximum (or a small number of localized extrema) in the scalp map.

However, topography-matching alone was not sufficiently reliable for automated detection. Across recordings, the apparent focal location could shift, and some components exhibited multiple focal regions or a less stereotyped spatial footprint (Figure 8). Because the location and exact shape of the DIN scalp projection could not be predicted consistently enough to support a generalizable rule, we did not pursue a purely topography-driven classifier. Instead, topography was retained as a supportive cue during manual review, while the automated decision metric relied primarily on the frequency-domain harmonic signature, which was sufficient for our intended use in routine artifact identification.

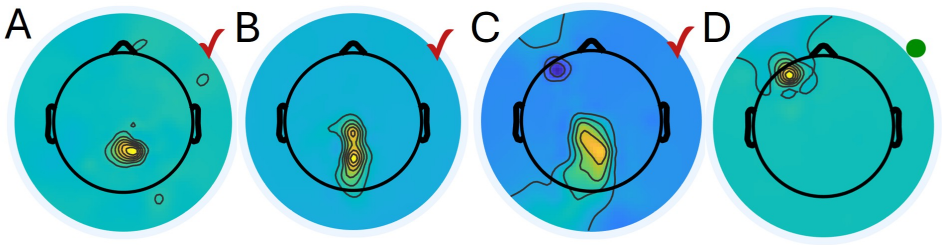


Figure 8: **Representative DIN ICA topographies across recordings.** Each scalp map shows an ICA component identified as DIN contamination in a different recording. *Panels A and B*: Components exhibit the expected focal DIN scalp distribution centered near Pz. *Panel C*: A dual-focal projection with one focus near Pz and a second near Fp1. *Panel D*: An atypical DIN component concentrated near Fp1 rather than the usual posterior focus.

CHAPTER 5

Example Workflow: ICA Removal of EKG and DIN Components

This section walks through the tutorial script `example.m` in the `EKG-DIN Artifact Detection Module`. This walkthrough highlights important variables and configuration steps. It does not explain every line of code, but instead focuses on the elements that users are expected to adjust when applying the Module to their own datasets.

The script shows how to:

1. Specify the data location and load an example EEG file.
2. Configure parameters and output folders.
3. Obtain or compute the ICA unmixing matrix W.
4. Run the automatic EKG and DIN artifact detections and interact with the manual review UI.
5. Perform extra inspection of ICA sources.
6. Reconstruct data from ICA space.
7. Visualize the effect of removing selected sources.



To run the tutorial, open **example.m** in MATLAB, ensure that the **EkgDinArtifactDetection** folder is on your MATLAB path. Example:

```
addpath(genpath('SENSI-EEG-Preproc-ekg-din-artifact-main'))
```

- Where, "SENSI-EEG-Preproc-ekg-din-artifact-main" represents the folder containing the code base.
- Run the script section-by-section.
- If you do not change the folder paths (explained below), make sure you are located in the code base root folder when running the **example.m** script.



The example will download the sample files (e.g., `data1.mat`, `data1_W.mat`, `data2.mat`, `data2_W.mat`, ...) to a folder of your choice. Each file will contain, at minimum:

- **Data files** (e.g., `data1.mat`)
  - `xRaw`: filtered EEG data (2D)
  - `fs`: sampling rate (Hz)
  - `badCh`: indices of channels excluded from `xRaw`
- **ICA unmixing matrix file** (e.g., `data1_W.mat`)
  - `W`: ICA unmixing matrix

5.1 Step 1: Setup

Paths, Filenames, and Artifact Options

The first section of `example.m` sets the data location, selects which example file to load, specifies where to save figures, and defines all parameters used by `autoDetectDIN()` and `autoDetectEkg()` via the `dinOpts` and `ekgOpts` structures.

```
%% 1. SETUP AND INPUT PARAMETERS

% FILE AND PATH CONFIGURATION
dataFolder = 'ExampleData\';  % Directory containing data files
INFO.figDir = 'Figures\';    % Output directory for saving figures
INFO.saveFigs = 1;            % 1 to save review figures, 0 otherwise

% Choose one example file:
%   'data1' : Example dataset 1
%   'data2' : Example dataset 2
%   'data3' : Example dataset 3
fileName = 'data1'; % <-- EDIT THIS

% Electrode locations file
locfile = 'EGI-9_18-124.sfp'; % name of electrode locations file

%%%% ARTIFACT DETECTION PARAMETERS %%%%
% These thresholds control the automated component flagging process.

% DIN (Digital Input) Parameters
dinOpts.T = 1;     % Din repetition period in seconds. Default: 1 sec
dinOpts.fmin = 10; % Minimum frequency for detection (Hz)
dinOpts.fmax = 50; % Maximum frequency for detection (Hz)

% EKG (Cardiac) Parameters (default values already loaded)
ekgOpts.fmin = .8;        % Lower bound of cardiac rhythm band (Hz)
ekgOpts.fmax = 2;         % Upper bound of cardiac rhythm band (Hz)
```



Key points:

- **`dataFolder`** and **`INFO.figDir`** may be updated to prevent path errors.
- **`fileName`** selects which dataset is used in the tutorial.



- The script assumes an EGI-style 128-channel recording (129 channels minus the reference channel and any bad channels).

These values are reasonable starting points for 128-channel EGI data. Users can tune them based on their own needs.

Guidelines for Choosing Parameters. The parameters in this section control how `autoDetectDIN()` and `autoDetectEkg()` score ICA components using frequency-domain features. Below are practical guidelines for tuning these values to your recordings.

- `autoDetectDIN()` parameters (`dinOpts`)
  - **Repetition period (`dinOpts.T`).** `dinOpts.T` sets the expected repetition period (s) of the DIN artifact. This determines the spacing of harmonic peaks in the spectrum via $f_0 = 1/T$. For example, `T=1` corresponds to 1 Hz spacing. If your DIN trigger repeats at a different rate, update `T` accordingly.
  - **Detection band (`dinOpts.fmin`, `dinOpts.fmax`).** These bounds define the frequency range used to evaluate the comb-like harmonic structure. Choose `fmin` high enough to avoid slow drifts and low-frequency physiological activity, and choose `fmax` to include multiple harmonics while avoiding frequency regions dominated unrelated noise. With `T=1`, a 10–50 Hz band typically captures many harmonics and yields stable scoring.
  - **Common tuning adjustments.**
    - If DIN harmonics are visible but scoring is weak, increase `dinOpts.fmax` (more harmonics) or decrease `dinOpts.fmin` slightly (include lower harmonics).
    - If false positives occur due to broadband activity, raise `dinOpts.fmin` and/or reduce `dinOpts.fmax` to a cleaner band.
- `autoDetectEkg()` parameters (`ekgOpts`)
  - **Cardiac rhythm band (`ekgOpts.fmin`, `ekgOpts.fmax`).** These bounds define the expected heart-rate band for the fundamental cardiac rhythm (and associated harmonics). Reasonable defaults for resting adults are approximately 0.8–2 Hz (≈48–120 bpm).
  - **Common tuning adjustments.**
    - If higher accuracy is required, the search band can be narrowed to reflect a more precise expected heart-rate range.

* If subjects are tachycardic or recordings are collected during movement, increase `ekgOpts.fmax`.
* If recordings are from sleep or very relaxed states, `ekgOpts.fmin` may be reduced slightly.

5.2 Step 2: Download and Load Data

The second section downloads the example data if it is not already included in the **ExampleData** folder. It also converts the electrode locations file into a struct and removes any channels that were previously excluded.

Outputs

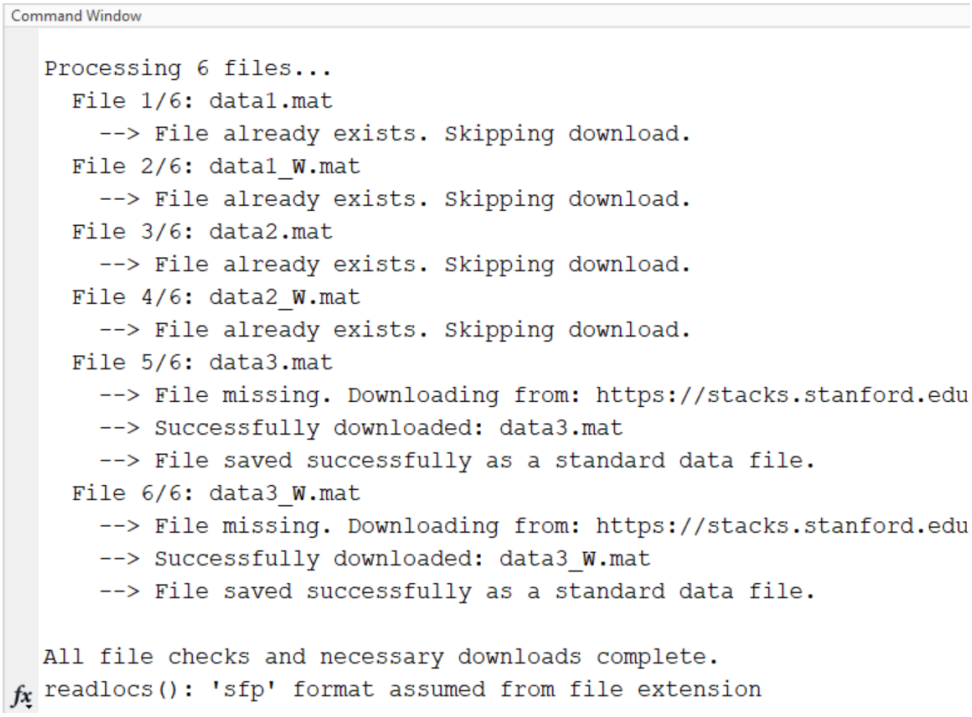


Figure 9: **Downloading files**. Section 2 will ensure the example files are contained in the **"dataFolder"** directory. If not present, it will download them to the defined directory.

If there is an error while downloading the data, you may do the following:

1. Comment out `"getExampleData_ekgdin(dataFolder);"`
2. Download the sample data[a]
3. Place sample data in the **ExampleData/** folder, or update the **dataFolder** variable with your desired data folder.

[a] https://purl.stanford.edu/xd818jt7842

5.3 Step 3: ICA Decomposition (Obtain W and Compute `xICA`)

The artifact detectors operate on ICA component time series, so the sensor-space data `xRaw` must first be transformed into ICA space. This requires an ICA unmixing matrix W, which maps sensor-space signals to independent components:

$$\texttt{xICA} = W\,\texttt{xRaw}.$$

Two supported paths to obtain W.

1. **Load a precomputed W (tutorial convenience).** The example script first attempts to load a cached unmixing matrix from:

   $$\texttt{wPath} = \texttt{ExampleData/[INFO.fileName_W.mat]}.$$

   This avoids recomputing ICA and makes the tutorial run quickly.

2. **Compute W from the current data (general use case).** If the cached file is not found, the script runs ICA directly:

   $$W = \texttt{doICA(xRaw)},$$

   then saves the result to `wPath` for future runs.

Using your own dataset. To run this workflow on user data, either provide a precomputed W, or allow `doICA()` to compute W from `xRaw`.

5.4 Step 4: Automated Detection and Interactive Review

This step performs automated candidate selection followed by interactive confirmation. Two UIs are launched in sequence:

1. **DIN review UI** (from `autoDetectDIN()` and `reviewDINArtifactUI()`),
2. **EKG review UI** (from `autoDetectEkg()` and `reviewEkgArtifactUI()`).

What to look for (practical criteria). Each row in the review UIs shows three complementary views of the same ICA component: *(i) topography*, *(ii) time-domain segment*, and *(iii) frequency-domain magnitude spectrum*. Use all three jointly; no single panel is definitive.

Candidates start flagged for removal (red). Clicking a row toggles it to keep (green). The review is finalized by pressing *Done*. If `INFO.saveFigs = 1`, the final UI state is saved when *Done* is clicked.

EKG candidates (`reviewEkgArtifactUI`). ICA components capturing EKG artifacts typically show the following:

- **Topography:** a broad or smoothly varying scalp projection. Orientation and distribution may vary across recordings, so topography should be used as confirmation rather than a strict rule.
- **Time-domain:** repeated cardiac beat morphology with short, sharp peaks occurring at a roughly regular cadence. Beat spacing may vary mildly (heart-rate variability), but the pattern should remain clearly repetitive within the displayed segment.
- **Frequency-domain:** a dominant low-frequency fundamental (f_0) within the cardiac band ($\sim$ 0.8–2.0 Hz), accompanied by multiple integer harmonics that appear as repeated peaks ($f_1 = 2f_0,\ f_2 = 3f_0,\ f_3 = 4f_0$). The key cue is a *harmonic series* rather than a single isolated peak.

DIN candidates (`reviewDINArtifactUI`). ICA components capturing DIN artifacts typically show the following:

- **Topography:** often a focal region (hotspot-like) around Pz and non-dipolar.
- **Time-domain:** sharp pulses occurring at fixed intervals. These events reflect externally triggered digital input (DIN) pulses.
- **Frequency-domain (primary cue):** a pronounced harmonic comb at integer multiples of the DIN base frequency $f_0 = 1/T_{\mathrm{DIN}}$.

Practical rule of thumb. For EKG, require a clear beat-like waveform *and* a harmonic series. For DIN, require a clear harmonic comb with spacing f_0.

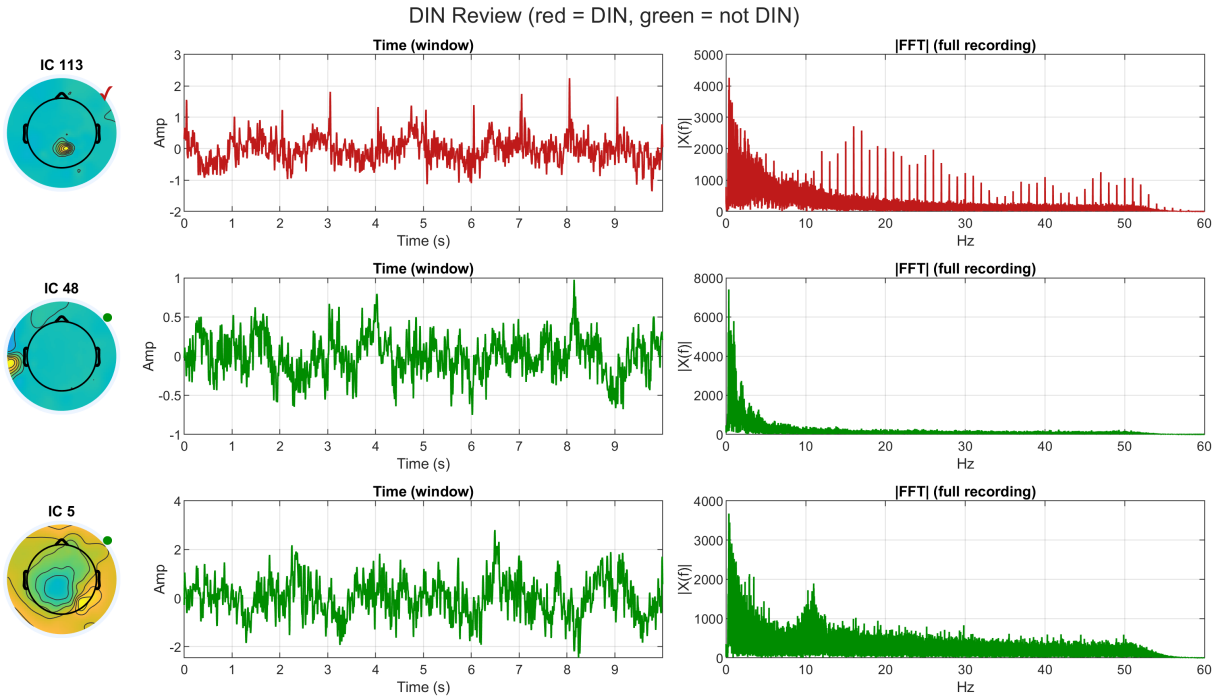


Figure 10: **DIN interactive review interface.** Component 113 is flagged as a DIN artifact, while Components 48 and 5 are shown as additional possible candidates. Based on the combined evidence from topography, time-domain activity, and the frequency spectrum, only Component 113 exhibits the characteristic DIN structure. The default selections are therefore retained, and the review is finalized by pressing **Done**.

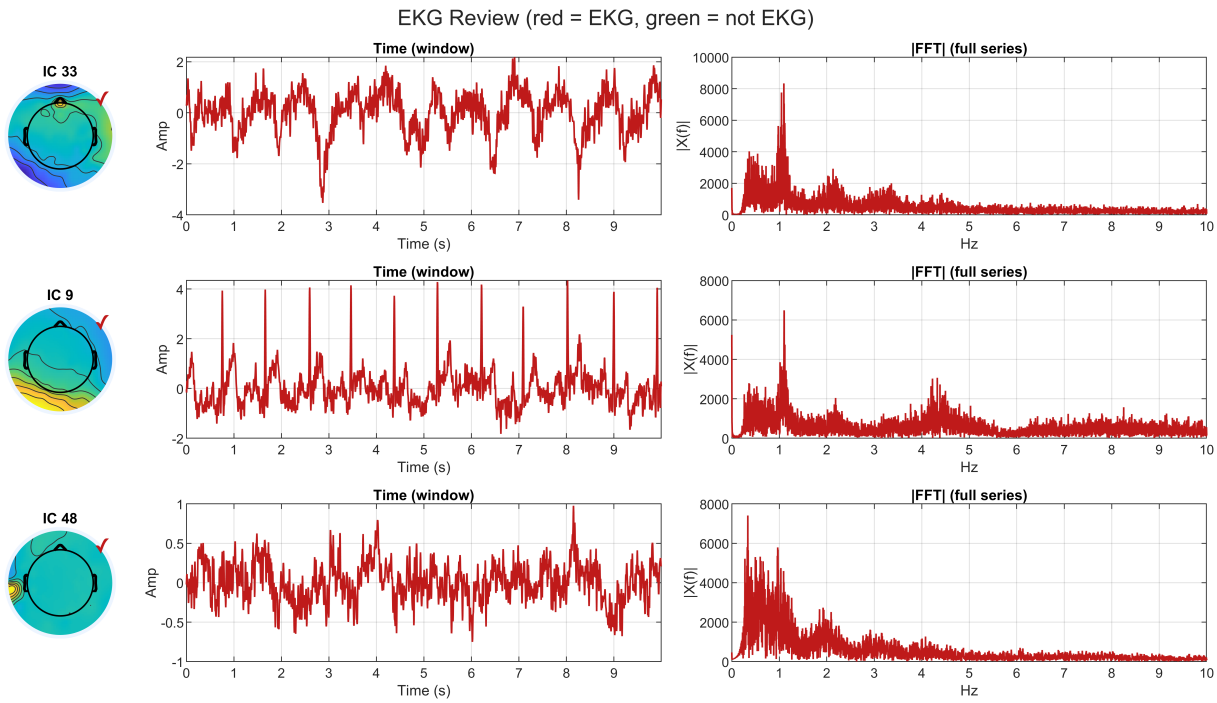


Figure 11: **EKG interactive review interface (before).** All components shown are initially flagged as EKG candidates due to their spectral structure. Component 9 exhibits the clearest cardiac signature across all three panels: a beat-like waveform in the time series, an expected topography, and a dominant fundamental near the heart-rate band with multiple harmonics in the spectrum. Component 33 is less obvious in the displayed time window, but still consistent with cardiac contamination. Cardiac activity is not always captured by a single ICA component. Different portions of the cardiac cycle (e.g., sharp QRS-like deflections versus slower wave components) may separate into different ICs. As a result, multiple ICs may exhibit cardiac periodicity, with differing waveform morphology and topographic projection. For Component 33, the strongest evidence is spectral: a pronounced peak near ~1.1 Hz followed by clear harmonics (matching component 9). Its time-domain deflections appear weaker in this segment (and may be polarity-inverted), consistent with a slower cardiac-related component rather than sharp QRS-like transients. Component 48 shows weaker EKG indicators compared to Components 9 and 33.

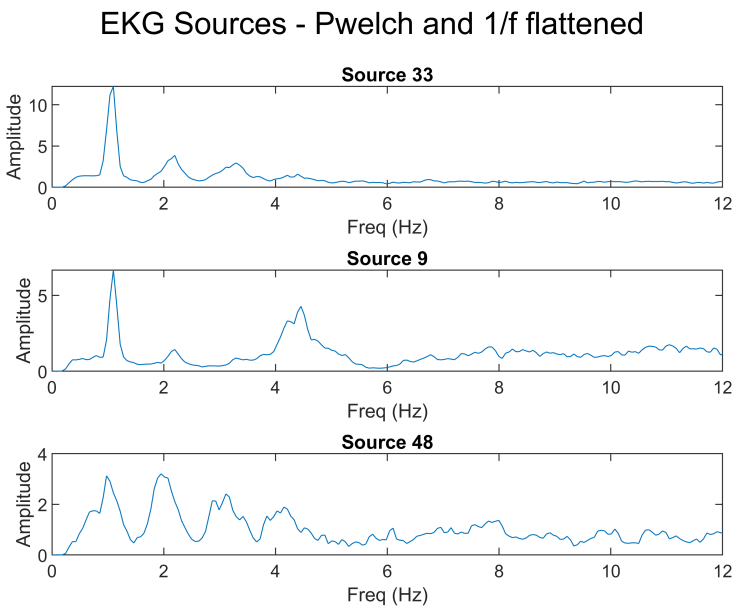


Figure 12: **EKG component spectra.** This helper plot shows a smoothed Welch power spectrum for each candidate component, improving visibility of the fundamental heart-rate peak and its harmonics. These spectra are the basis of the harmonic local-SNR scoring used by `autoDetectEkg()`. All three components exhibit some cardiac influence, but Components 9 and 33 show the most prominent cardiac signature, including a strong peak near ~1.1 Hz and its harmonics.

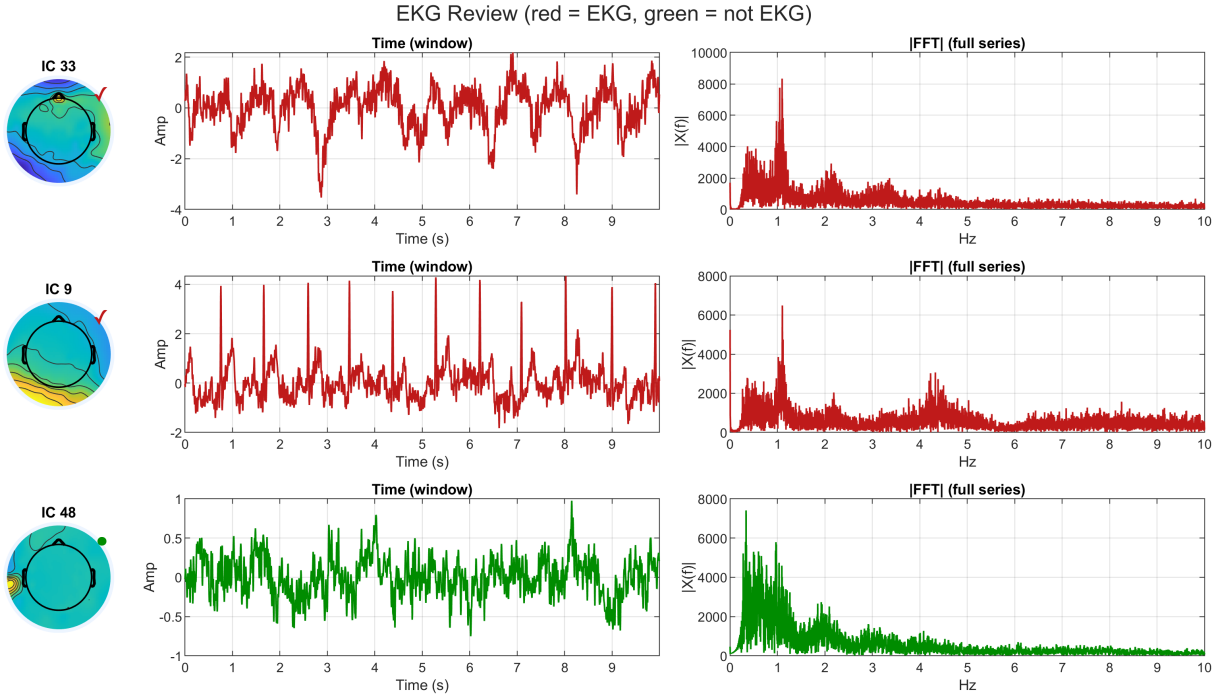
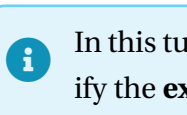


Figure 13: **EKG interactive review interface (after).** Based on the evidence summarized in Figure 11 and Figure 12, Components 9 and 33 are marked for removal, while Component 48 is retained.

5.5 Step 5: Review of “Other” Components

The example script includes an optional review step (`reviewOtherUI()`) to demonstrate how additional ICA components may be inspected after EKG/DIN detection. This step is not part of the intended EKG/DIN detection workflow and is included only for tutorial completeness. All independent components will appear as green and the user may reject them as needed. Explanation of what components to keep or retain is outside of the scope of this manual.



In this tutorial, we are only reviewing the first 10 components. The user may modify the **example.m** script to review more components.

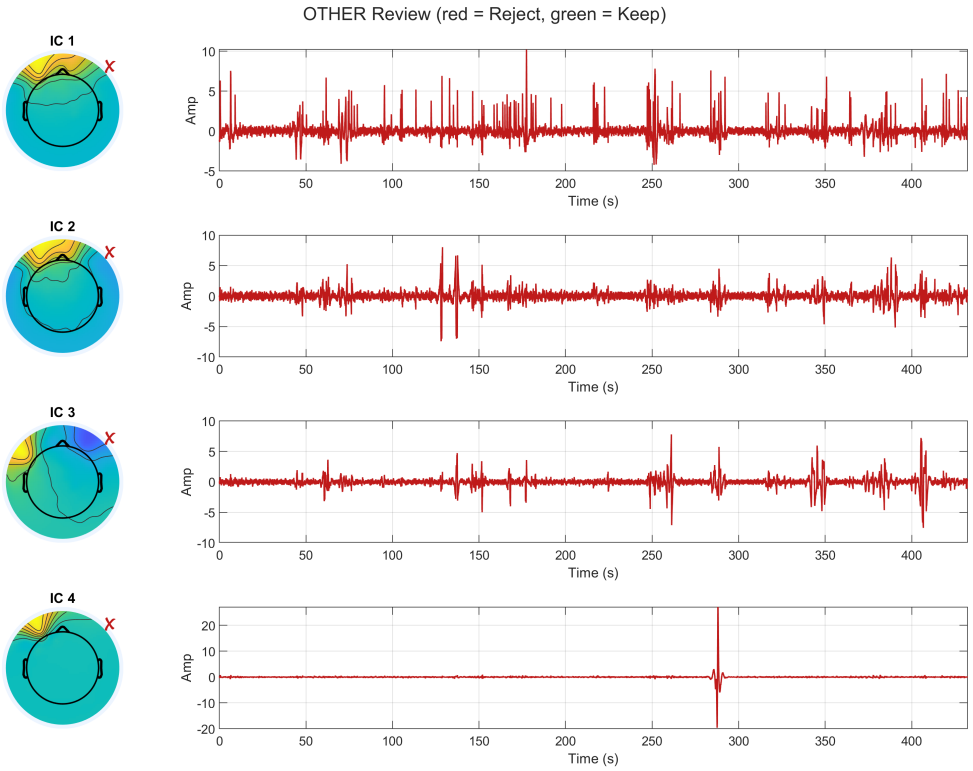


Figure 14: **Other ICA components (Page 1).** All components in this page were tagged for removal.

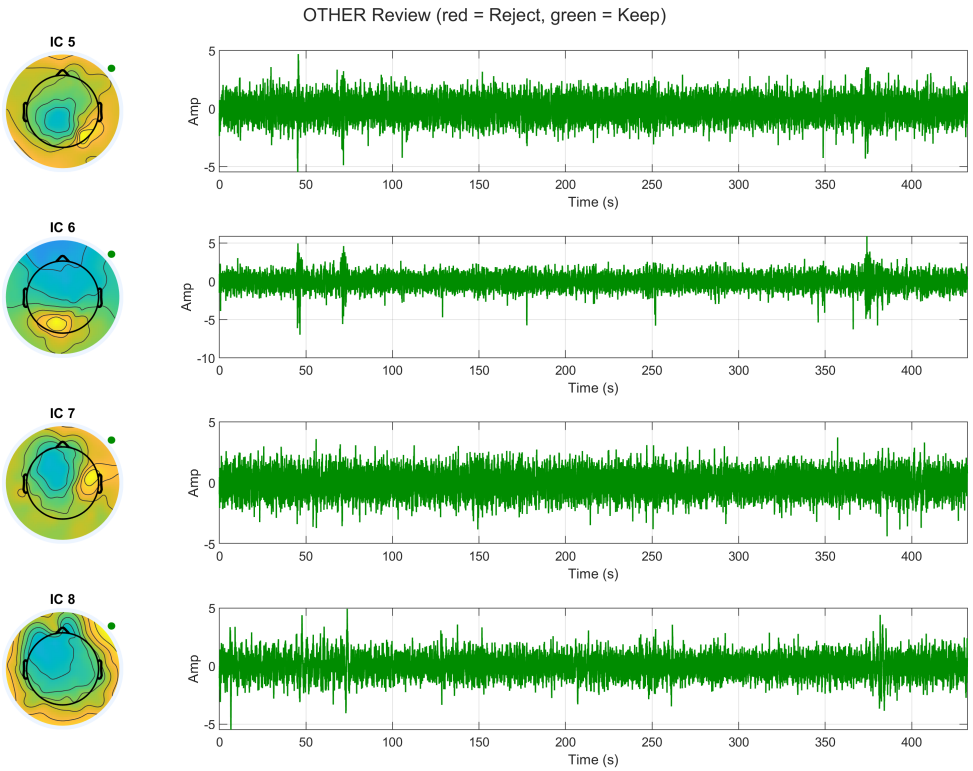


Figure 15: **Other ICA components (Page 2).** All components in this page were kept.

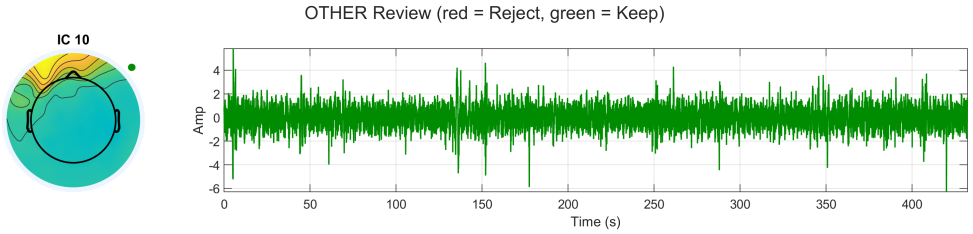


Figure 16: **Other ICA components (Page 3).** This component was kept.

5.6 Step 6: Artifact Removal and Signal Reconstruction

After interactive review, the final component lists are combined:

$$\texttt{rmSrc} = [\texttt{ekgSrc}; \texttt{dinSrc}; \texttt{otherSrc}].$$

Artifact components are removed by zeroing them in ICA space:

$$x_{\text{ICA}}(\texttt{rmSrc}, :) \leftarrow 0,$$

and the cleaned sensor-space signal is reconstructed:

$$x_{\mathrm{Cl}} = W^{-1} x_{\mathrm{ICA}}.$$

5.7 Step 7: Quality Control — Before and After Plots

The final block produces a qualitative before/after comparison between:

- **Before:** `xRaw`
- **After:** `xCl`

Interpretation. The before/after figures are not intended to "prove" correctness. They are a sanity check that large structured artifacts are reduced and that no obvious recording-level failures remain after ICA component removal.

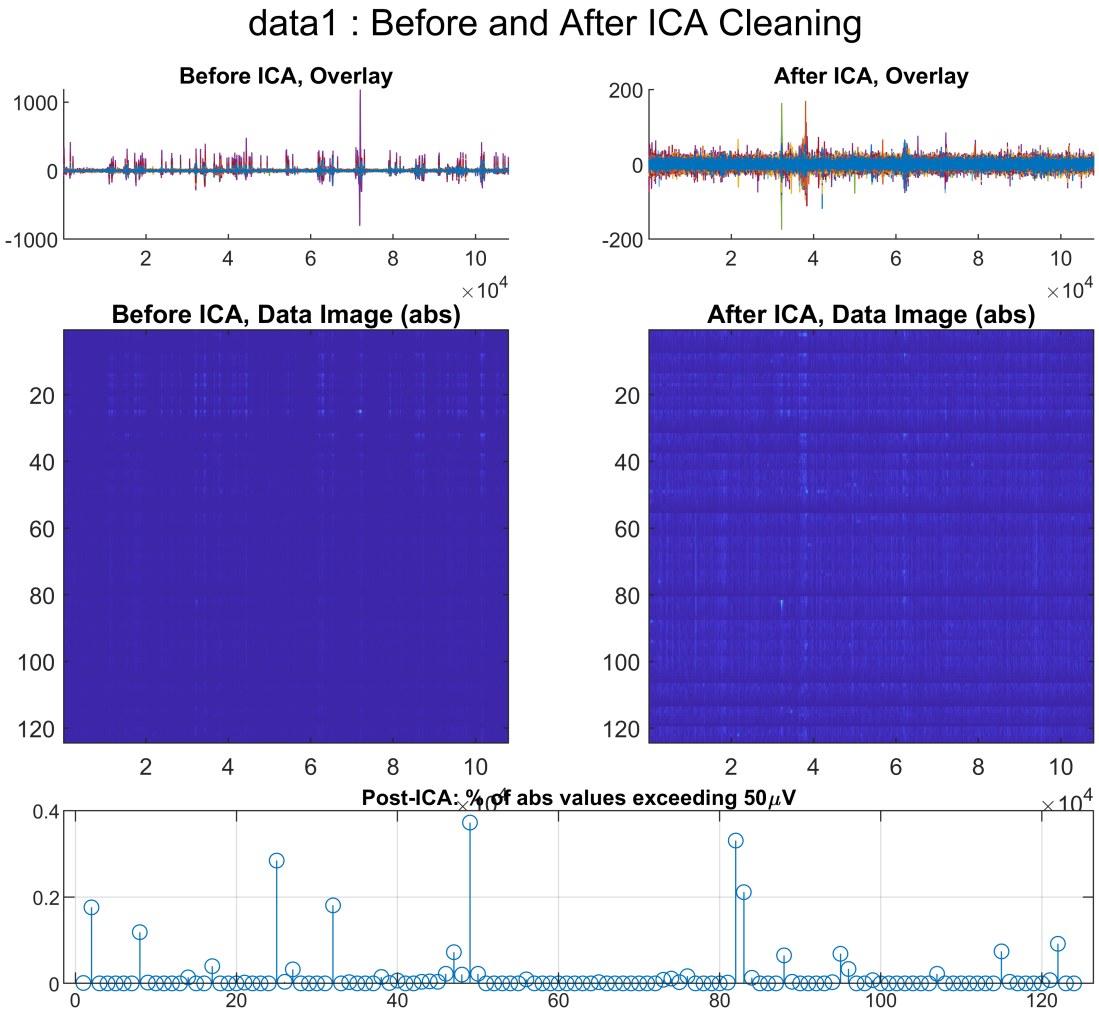


Figure 17: **data1 — Before vs. After artifact removal.** The script displays: (i) overlay plots, (ii) a data-image view of absolute magnitude, and (iii) a post-ICA per-channel outlier-rate summary using thresholds `INFO.recUVThresh` and `INFO.recPctThresh`.

CHAPTER 6

Additional Illustrative Examples

Overview

This chapter provides focused illustrative examples intended to improve user interpretation of the DIN and EKG review interfaces. Unlike the walkthrough in `example.m`, the examples here emphasize only the decision cues. Each example follows the same structure: (i) what was flagged automatically, (ii) what the user should prioritize in the UI panels, and (iii) the final keep/remove decision.

No new code is introduced here: each example uses the tutorial script `example.m`, changing only the `fileName` variable to `data2.mat` or `data3.mat` and re-running the workflow described in Section 5.

6.1 Example 2

Data summary

Data used

- Dataset: `data2.mat & data2_W.mat`
- Channels: **113** scalp channels after removal of bad channels (EGI-style)
- Sampling rate: **250 Hz**
- Notes:
  - **Manual interaction required.**
  - Multiple bad channels removed in previous steps.

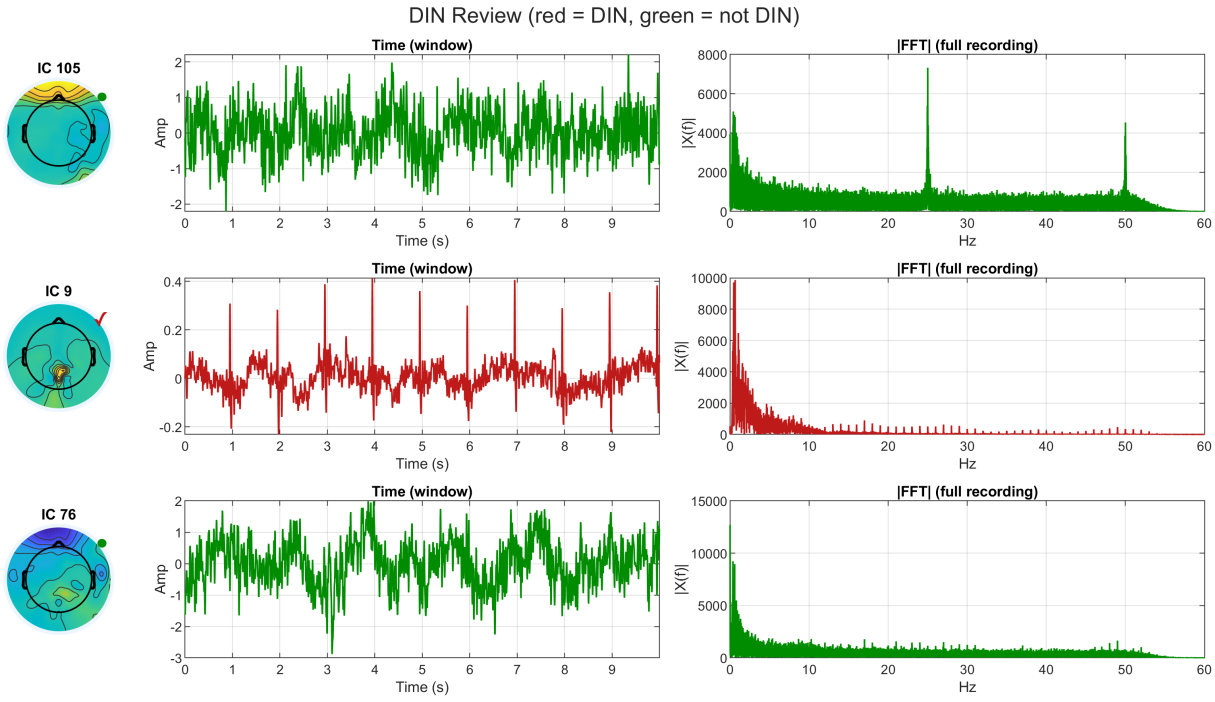


Figure 18: **DIN review (Example 2).** Components 105 and 9 were initially flagged for removal *(marked as red)*. Component 105 shows elevated spectral peaks near ~25 Hz and ~50 Hz, but does not exhibit a consistent DIN harmonic-comb signature and is retained *(change mark to green)*. In contrast, Component 9 shows the expected DIN pattern across panels: a localized topography (centered near Pz), a stereotyped waveform with regularly repeating transients in the time segment, and a frequency-domain comb consistent with the configured repetition period. Component 9 is therefore the only component marked for removal. Component 76 does not show DIN characteristics and it is retained.

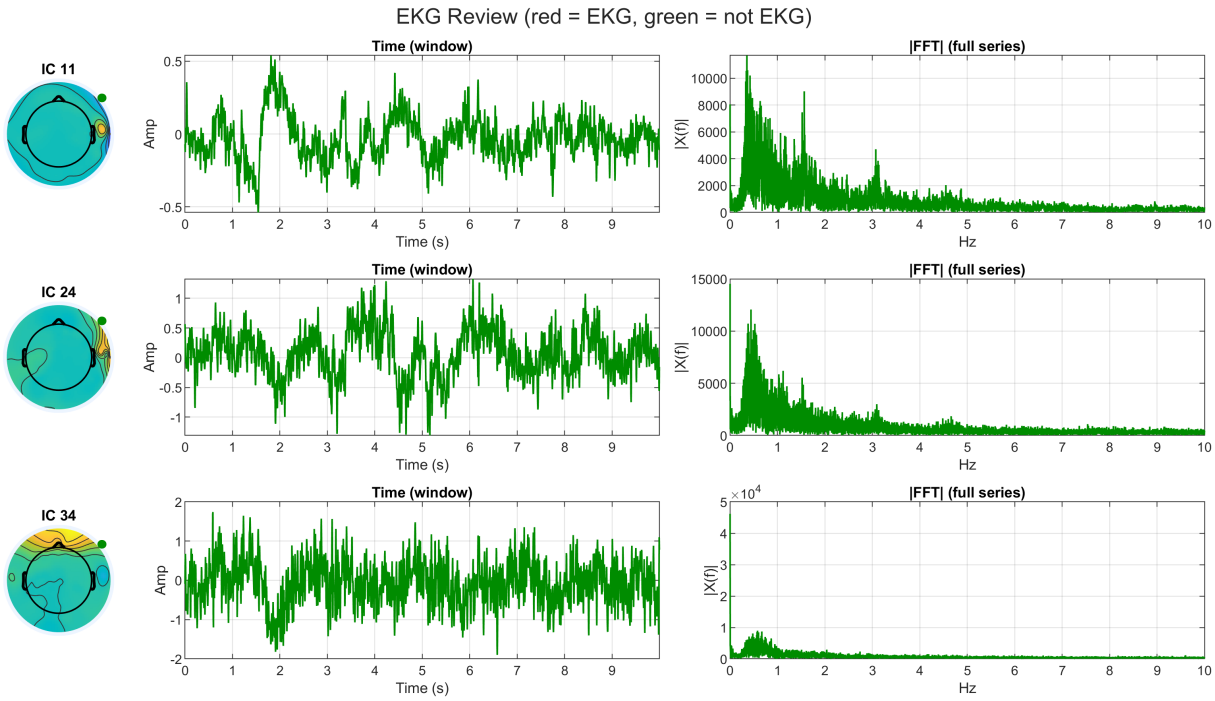


Figure 19: **EKG review (Example 2).** Components 11 and 24 were initially flagged for removal *(marked as red)* based on spectral structure within the physiological heart-rate band. For example, Component 11 shows a peak near ~1.5 Hz (approximately 90 bpm), denoted here as the putative cardiac fundamental (f_0), with additional energy near ~3 Hz ($2f_0$) and ~4.5 Hz ($3f_0$). However, visual inspection does not support an EKG classification: the displayed time segment does not show a beat-like waveform, and the scalp topographies are not consistent with typical cardiac projections. Moreover, true EKG components are typically dominated by the fundamental in the heart-rate band f_0. These candidates are therefore retained *(change marks to green)*. Component 34 does not show cardiac evidence and it is retained

Summary and final decisions.

- **DIN:** Components 105 and 9 were flagged. **Final:** remove Component 9; keep Component 105.
- **EKG:** Components 11 and 24 were flagged. **Final:** keep both (no components removed as EKG).

6.2 Example 3

Data summary

Data used

- Dataset: `data3.mat & data3_W.mat`
- Channels: **124** scalp channels (EGI-style)
- Sampling rate: **250 Hz**
- Notes:
  - **Manual interaction required.**

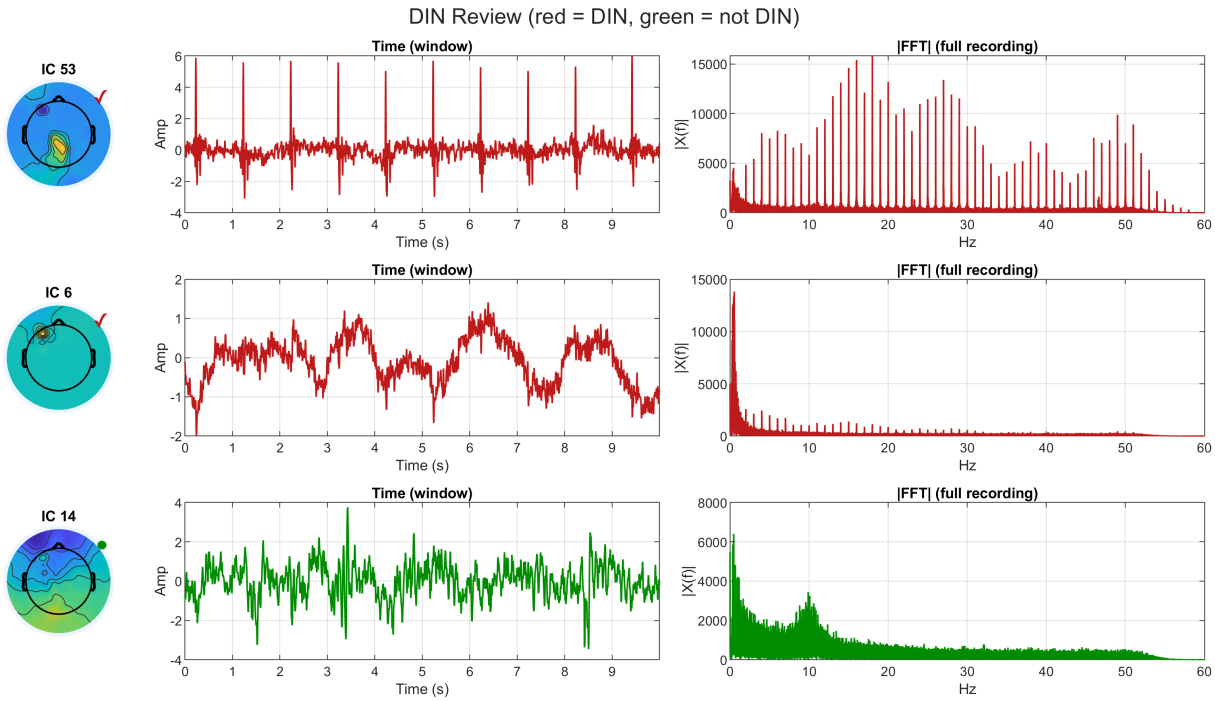


Figure 20: **DIN review (Example 3).** Component 53 was initially flagged for removal *(red)* and shows a clear DIN signature: a pronounced harmonic-comb spectrum and regularly repeating high-amplitude transients in the time segment. Its topography indicates strong projection near Pz, with additional weighting near Fp1, suggesting that the same interference pattern may project to multiple sensor regions. Component 6 was included as an additional review candidate *(green)*. It exhibits a weaker but still consistent comb structure in the spectrum and small-amplitude repeating transients in the time domain. Its topography shows a focus near Fp1, consistent with a complementary spatial projection relative to Component 53. Based on the combined evidence, Component 53 and 6 are confirmed as DINs and marked for removal. Component 14 does not show DIN characteristics and it is retained

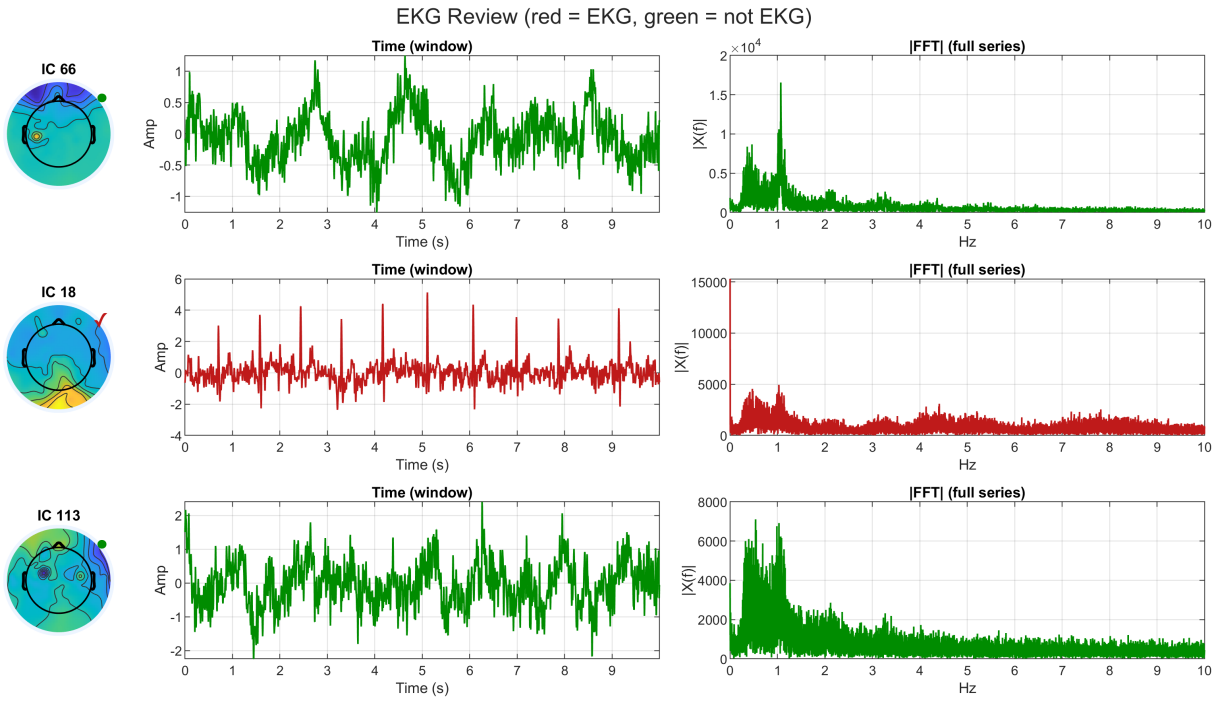


Figure 21: **EKG review (Example 3).** All components shown were initially flagged for removal *(red)*. Component 18 exhibits the clearest EKG signature across panels: a beat-like waveform with large deflections in the time domain, and a dominant spectral peak near the physiological heart-rate band ($f_0 \approx 1.1$ Hz) with harmonics at ~ 2.2, ~ 3.3, and ~ 4.4 Hz. Its scalp projection is broad and smoothly varying, consistent with a cardiac source. Component 113 does not show convincing cardiac evidence in either the time-domain waveform or the frequency spectrum and is retained. Components 66 and 113 are more ambiguous. Their spectra contains peaks aligned to the same harmonic series as Component 18, but the topography of component 66 is more characteristic of eye movements. Moreover, the time-domain waveforms of components 66 and 113 do not contain readily visible EKG activity. Since these questionable components are both high-numbered (implying lower variance), we would likely decide to reject component 18 but not components 66 or 113.

Summary and final decisions.

- **DIN:** Component 53 was flagged. **Final:** remove Component 53 and 6.
- **EKG:** Components 66, 18, and 113 were flagged. **Final:** remove Component 18; keep Component 66 and 113.

CHAPTER 7

Appendix: Helper functions

This chapter documents helper functions used by the EKG and DIN artifact detection pipelines.

1. **Project-native helpers** (`Helper Functions/postICAartifacts`): functions written specifically for this project.
2. **External utilities** (`Helper Functions/External Functions`): function(s) originating from an external toolboxes, included here to support a specific computational step.

Helper functions are documented in alphabetical order. For each helper function, we provide the syntax, a brief narrative description of its inputs and outputs, and its usage by other user-called and/or helper functions.

Notation. Throughout this chapter, `xICA` denotes ICA component time series arranged as $[nComp \times nSamp]$, and `W` denotes the ICA unmixing matrix, such that `inv(W)` yields component scalp maps as columns.

7.1 Project-native helpers

This subsection documents helper functions authored specifically for this project. These functions either (i) compute core diagnostic metrics used by the EKG/DIN detectors, or (ii) implement the interactive review interfaces used to confirm artifact classification.

`chanlocsFromFile()`

Syntax

```
chanlocs = chanlocsFromFile(loc_file)
```

Description. Loads an electrode-location file and returns a `chanlocs` structure compatible with `topoplot_new`. Converts a filename (e.g., `.sfp`) into a struct that can be reused for generating ICA scalp maps.

Usage. Used in **example.m** to load the electrode-location file specified by `locfile` and pass the resulting `chanlocs` to `topoplot_new` when generating ICA topomaps for DIN/EKG review.

`harmRatioFromSignal()`

Syntax

```
out = harmRatioFromSignal(x, fs)
out = harmRatioFromSignal(x, fs, opts)
```

Description. `harmRatioFromSignal()` computes an SNR-like frequency-domain metric that compares energy concentrated near expected harmonic frequencies of a repetition rate (f_0) against energy measured between harmonics.

Usage. Called by `autoDetectDIN()` to indicate the DIN structured harmonic contamination.

Fundamental definition. If `opts.T` is provided, the repetition period T (seconds) defines

$$f_0 = \frac{1}{T}.$$

If `opts.T` is empty, the function defaults to $T = 1$ second.

Output ratio. The metric returned is the ratio:

$$\texttt{out.ratio_linear} = \frac{\text{mean harmonic-band power}}{\text{mean inter-harmonic-band power}},$$

getExampleData_ekgdin()

Syntax

```
getExampleData_ekgdin(target_folder)
```

Description. `getExampleData_ekgdin()` downloads the example datasets used by `example.m` into a user-specified directory.

Usage. This function is called in **example.m** prior to loading `data1.mat`, `data2.mat`, or `data3.mat`. It ensures that all required example files are present on disk before the tutorial proceeds.

reviewDINArtifactUI()

Syntax

```
dinSrcFinal = reviewDINArtifactUI(W, xICA, fs, chanlocs, dinSrc, ...
              candidates, StartSec, NSec, figName, saveFigs, saveFolder)
```

Description. `reviewDINArtifactUI()` provides an interactive confirmation interface for DIN artifact classification. Components are shown in a tiled layout with one row per component:

[Topography] | [Time snippet] | [Frequency view].

Usage. Called after `autoDetectDIN()` to confirm candidate components. Components preflagged for rejection (`dinSrc`) begin in **red**, and review candidates begin in **green**. Clicking anywhere on a component's row toggles its membership (DIN vs keep). The review ends when the user presses *Done*. The final rejected set is returned as `dinSrcFinal`.

reviewEkgArtifactUI()

Syntax

```
ekgSrcFinal = reviewEkgArtifactUI(W, xICA, fs, chanlocs, ekgSrc, ...
              candidates, StartSec, NSec, figName, saveFigs, saveFolder)
```

Description. `reviewEkgArtifactUI()` provides an interactive confirmation interface for EKG artifact classification. Components are shown in a tiled layout with one row per component:

[Topography] | [Time snippet] | [Frequency view].

Usage. Called after `autoDetectEkg()` to confirm candidate components. Components preflagged for rejection (`ekgSrc`) begin in **red**, and review candidates (`candidates`) begin in **green**. Clicking anywhere on a component's row toggles its membership (EKG vs keep). The review ends when the user presses *Done*. The final rejected set is returned as `ekgSrcFinal`.

`reviewOtherUI()`

Syntax

```
keepIdx = reviewOtherUI(xICA, fs, compIdx, saveFigs, saveFolder, saveName)
```

Description. Interactive review interface for ICA components not automatically classified as EKG or DIN artifacts. All components are initialized as *keep* (green), and the user may toggle individual components to *reject* (red) based on visual inspection.

Usage. This function is not part of the intended artifact-detection workflow. It is included in `example.m` only to illustrate how ICA components may be reviewedand selectively removed in a broader preprocessing pipeline.



Important.

- Not required for execution of the artifact-detection module.
- Included solely for demonstration purposes in the example script.

7.2 External Utilities

This project incorporates a limited number of external utility functions originating from related toolboxes and academic laboratories. These functions support either (i) core artifact-detection computations, or (ii) example-script visualization and demonstration.

`computePerChannelPctOverThresh()`[7]

Syntax

```
[pctOverThresh, iMatrix] = computePerChannelPctOverThresh(xIn, thresh)
```

Description. Computes, per row (channel), the percentage of samples whose absolute magnitude strictly exceeds a specified threshold [12].

Usage. Called by `example.m` for post-ICA quality control. It computes, per channel, the percentage of samples exceeding INFO.recUVThresh, then compare against INFO.recPctThresh to warn about remaining high-voltage channels.

`doICA()`[7]

Syntax

```
W = doICA(x)
```

[7] https://github.com/blairkan/BKanMatEEGToolbox/tree/master

Description. Runs EEGLAB's `runica()` and returns the unmixing matrix `W`, which transforms data from channel space to ICA source space.

Usage. Called by `example.m` to compute the ICA unmixing matrix prior to DIN and EKG component detection.

`fillBadChRows()`[7]

Syntax

```
x124 = fillBadChRows(xIn, badCh)
```

Description. Reinserts previously removed bad-channel rows into a data matrix, filling those rows with `NaN` values so that the original channel indexing is restored [12].

Usage. Called by `example.m` after bad channels are removed prior to ICA, to restore the full channel layout (with NaNs in bad-channel rows) for reconstruction and plotting.

`plotEEGOverlay()`[7]

Syntax

```
plotEEGOverlay(x, vertLinePos)
```

Description. Plots a space-by-time data matrix as overlaid time-series traces in the current axis. Optionally overlays vertical reference lines (e.g., trial boundaries or event markers) [12].

Usage. Called by `example.m` for visual inspection of EEG channels and reconstructed signals.

`runica()`[8]

Syntax

```
[weights, sphere] = runica(data, varargin)
```

Description. Performs Independent Component Analysis (ICA) using the Infomax algorithm with natural gradient optimization and optional extended-ICA [13]. The function returns the unmixing matrix `weights` and sphering matrix `sphere`, such that:

```
W = weights * sphere
```

Usage. Called by `doICA()` within `example.m` to compute the ICA unmixing matrix.

[8] https://sccn.ucsd.edu/eeglab/

`topoplot_new()`[9]

Syntax

```
[h, Zi, grid, Xi, Yi] = topoplot_new(values, chanlocs, ...)
```

Description. Standalone scalp topography function adapted from EEGLAB's `topoplot()`. Generates 2-D interpolated scalp maps from channel values and electrode locations, with optional contour lines, electrode markers, and dipole overlays. Developed and distributed by the Parra Lab: https://parralab.org/

Usage. Used internally by the interactive review interfaces `reviewEkgArtifactUI()`, `reviewDINArtifactUI()`, and `reviewOtherUI()` to render scalp maps for spatial component visualization and artifact inspection.

[9] https://github.com/dmochow/SRC/blob/master/topoplot_new.m

CHAPTER 8

References

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